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(54) **BLARCAMESINE CO-CRYSTALS FOR THE MANUFACTURE OF PHARMACEUTICAL DOSAGE FORM**

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CPC **C07D 307/14** (2013.01); **C07B 2200/13** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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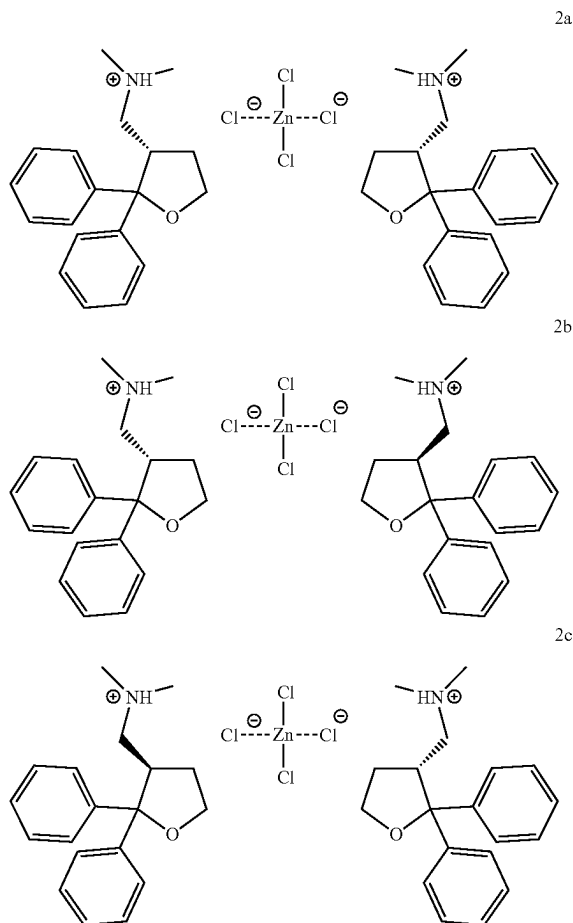
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(57) **ABSTRACT**

ZnCl₂ co-crystal of racemic(±)-blarcamesine hydrochloride of formula 2a, ZnCl₂ co-crystal of (S)-(-)-blarcamesine hydrochloride of formula 2b and ZnCl₂ co-crystal of (R)-(+)-blarcamesine hydrochloride of formula 2c, processes for preparation thereof and pharmaceutical preparation thereof and methods of treating neurodegenerative, neurodevelopmental diseases such as Alzheimer's disease, Parkinson's disease dementia and/or Rett syndrome.



19 Claims, 14 Drawing Sheets

Fig. 1.

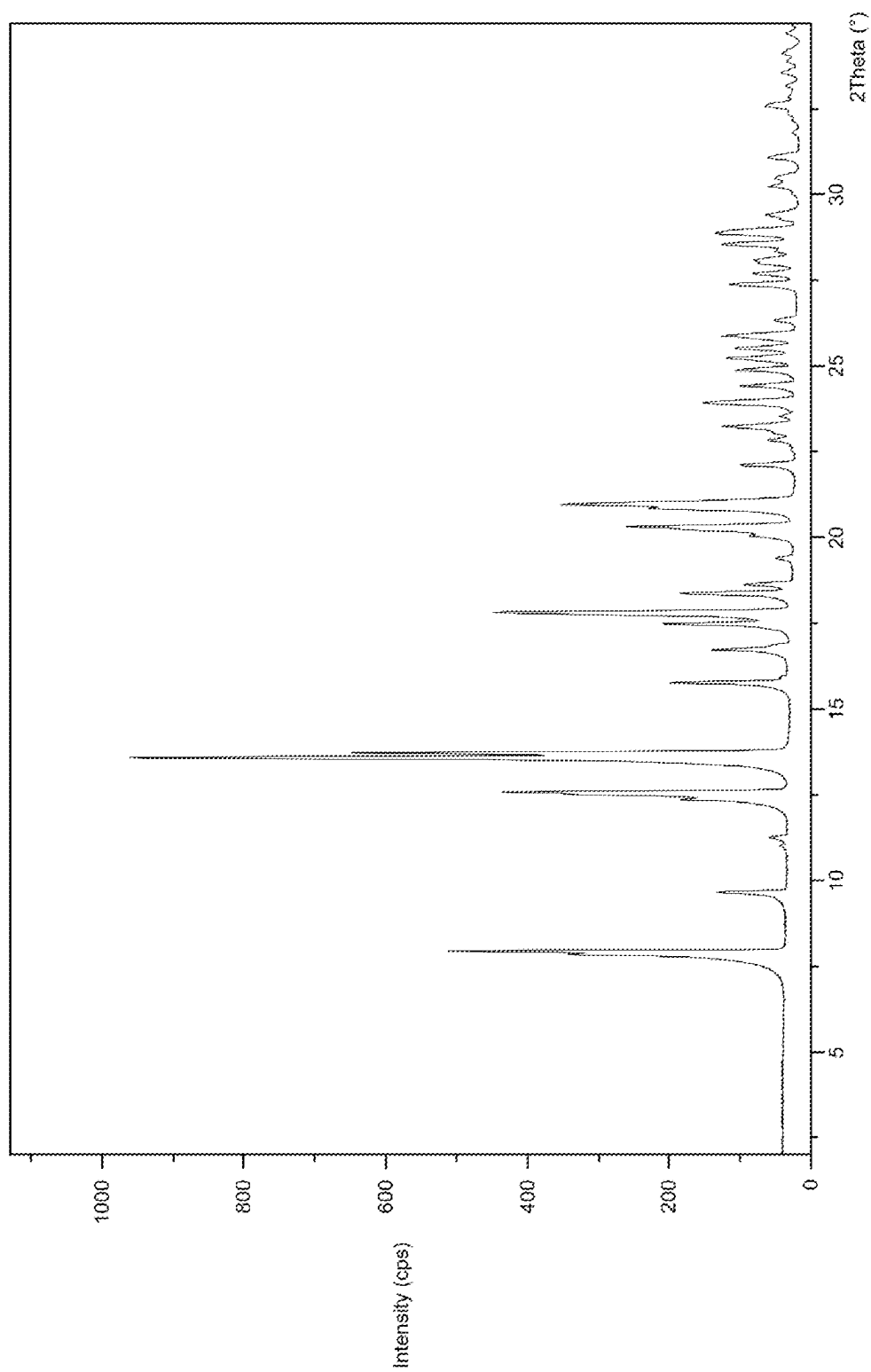


Fig. 2.

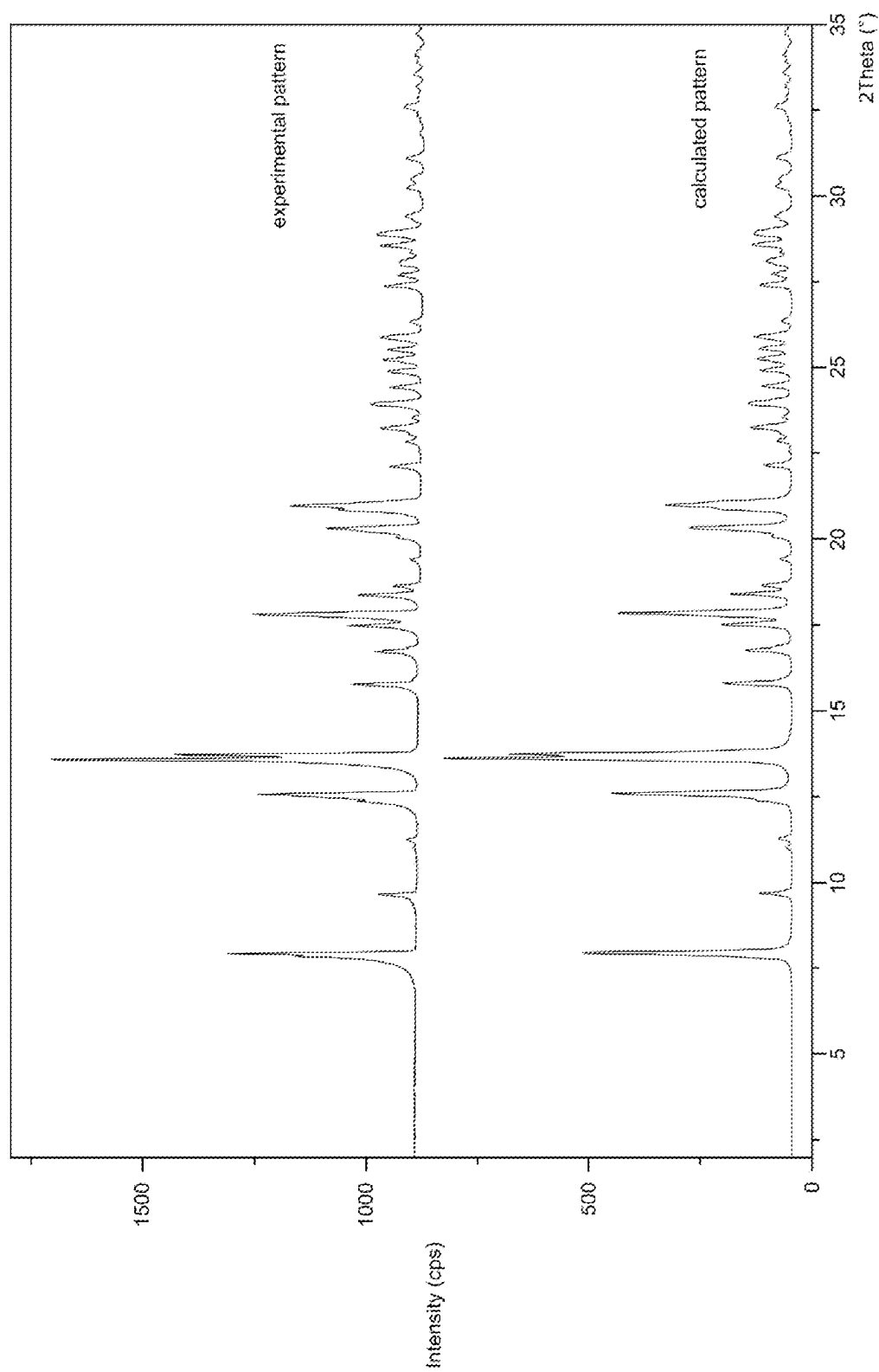


Fig. 3.

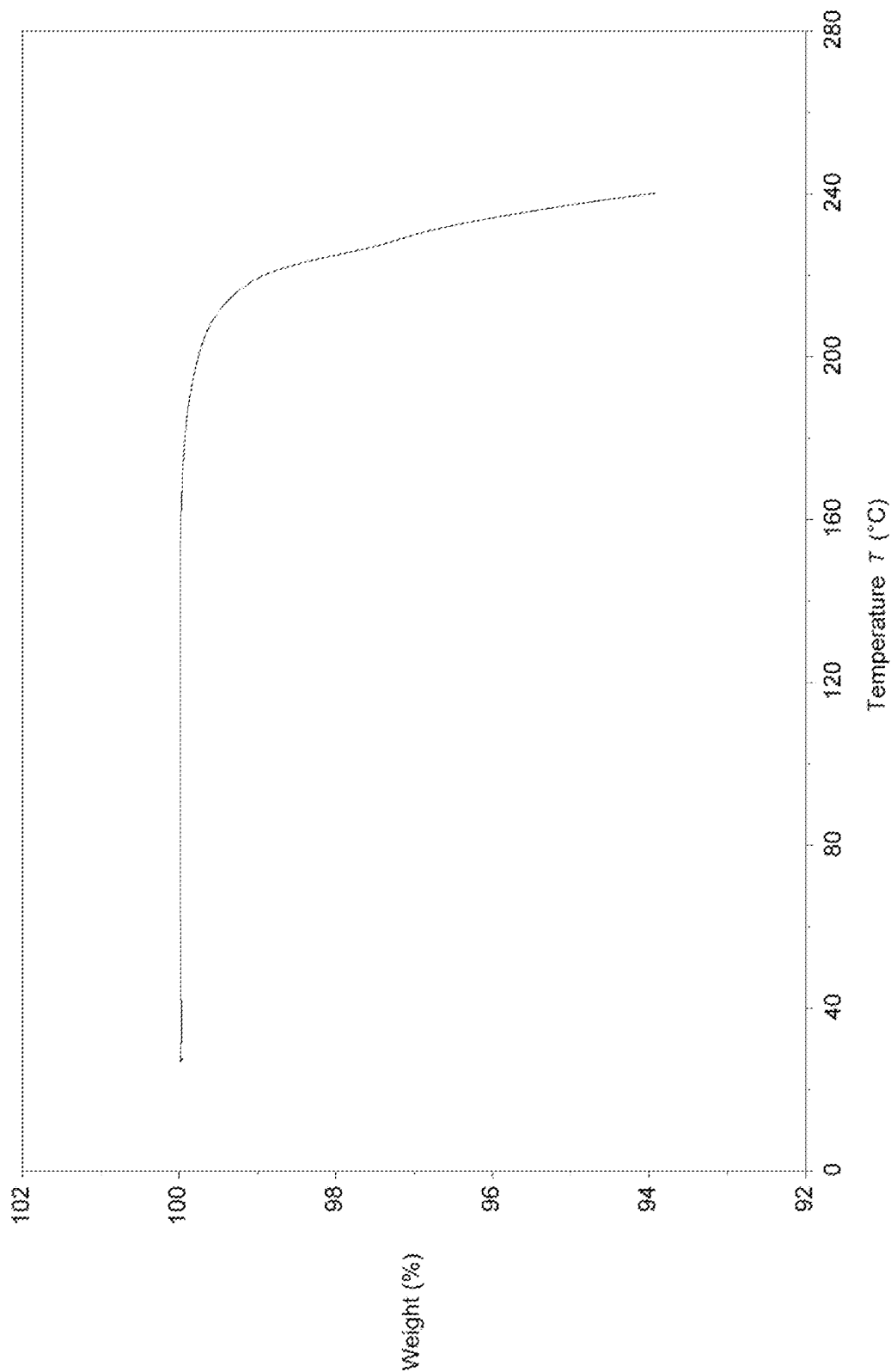


Fig. 4.

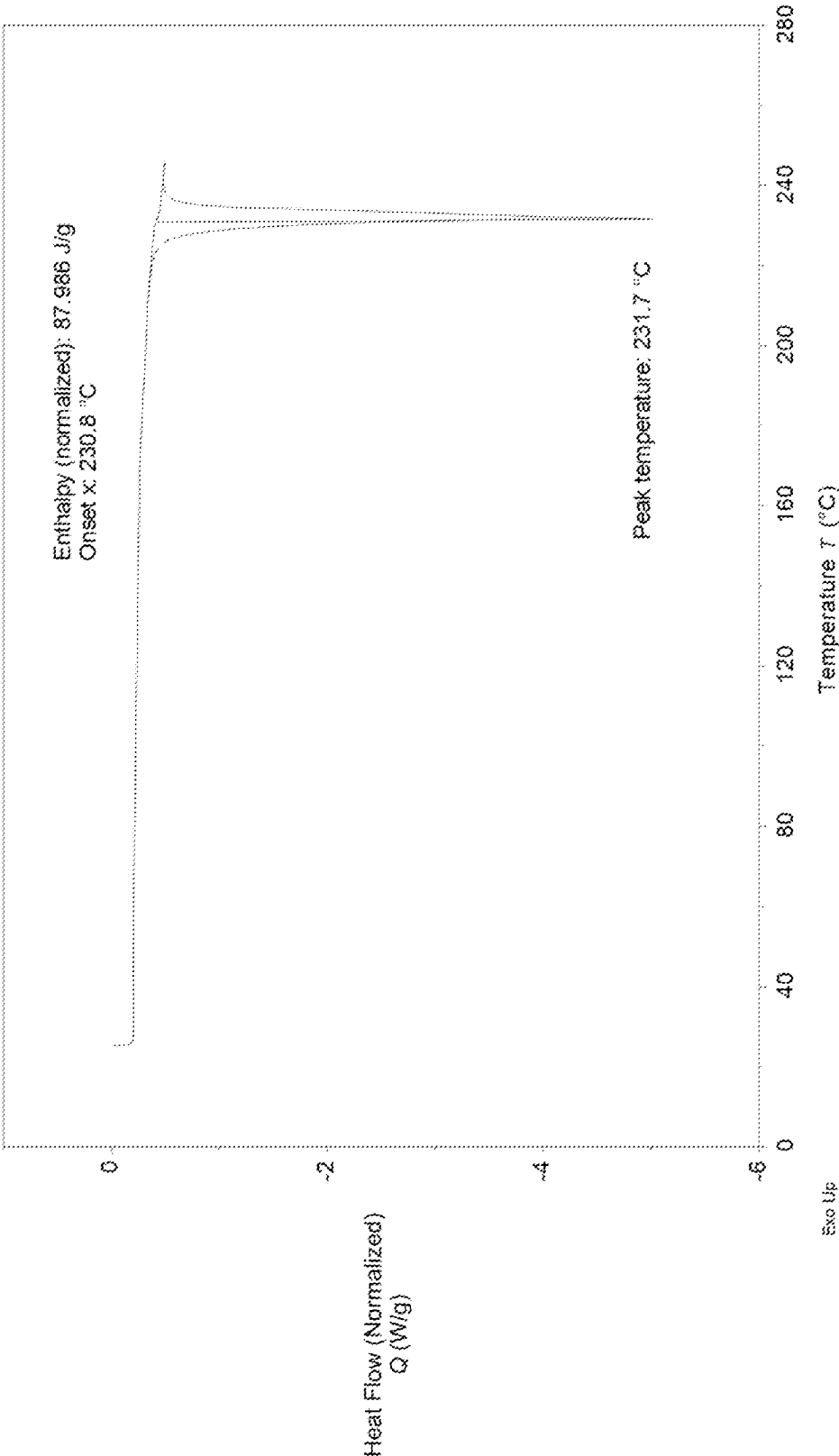


Fig. 5.

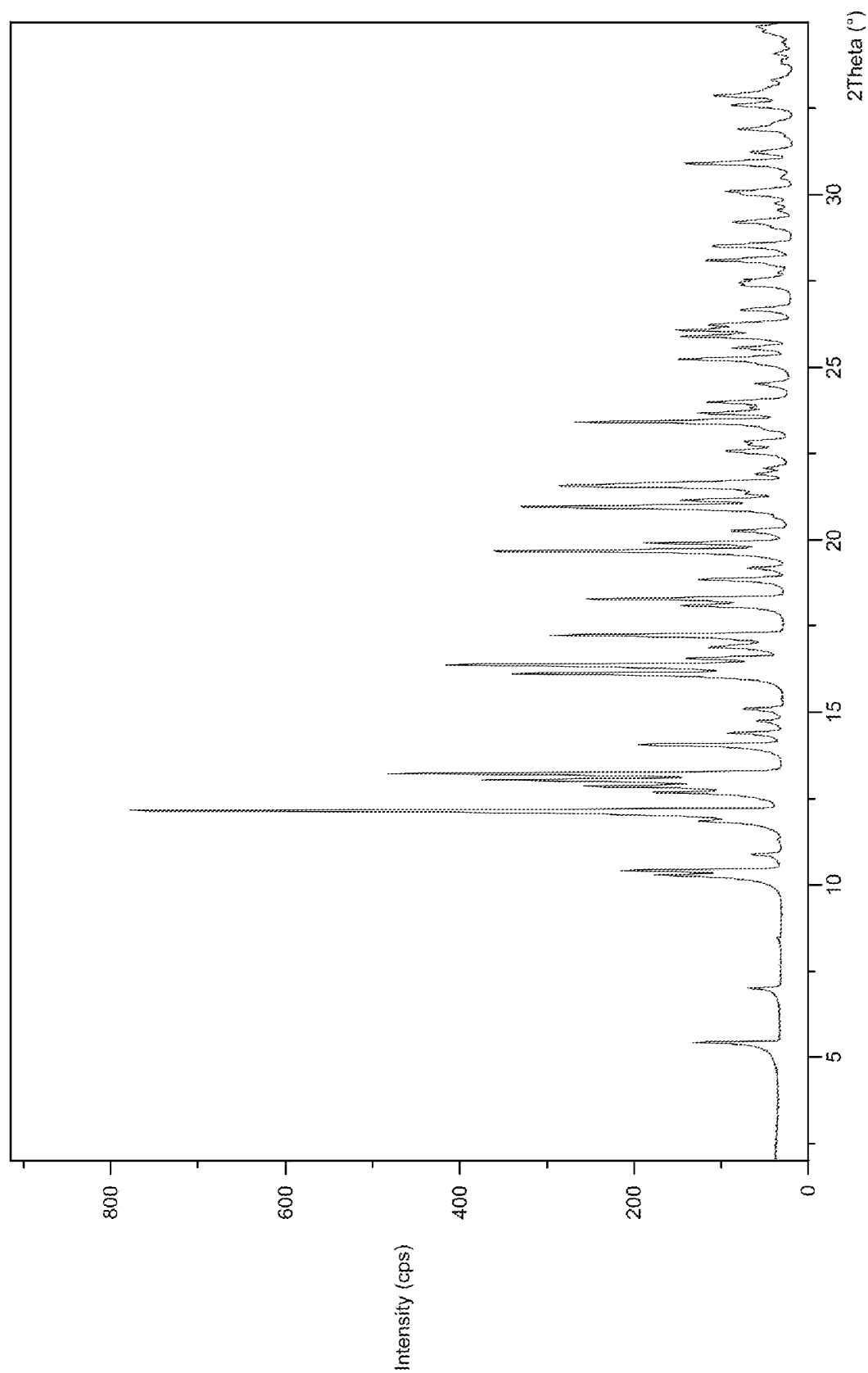


Fig. 6.

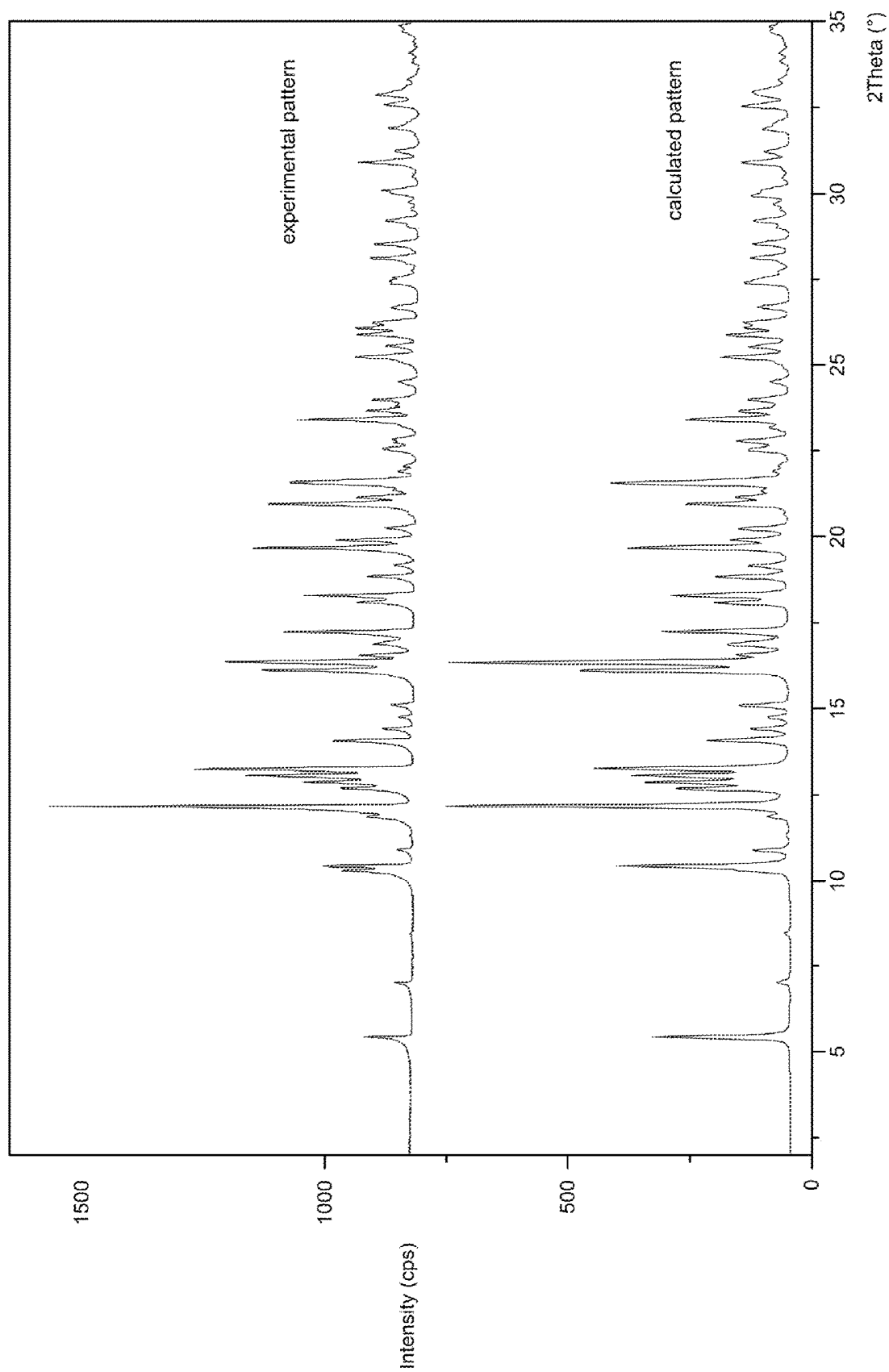


Fig. 7.

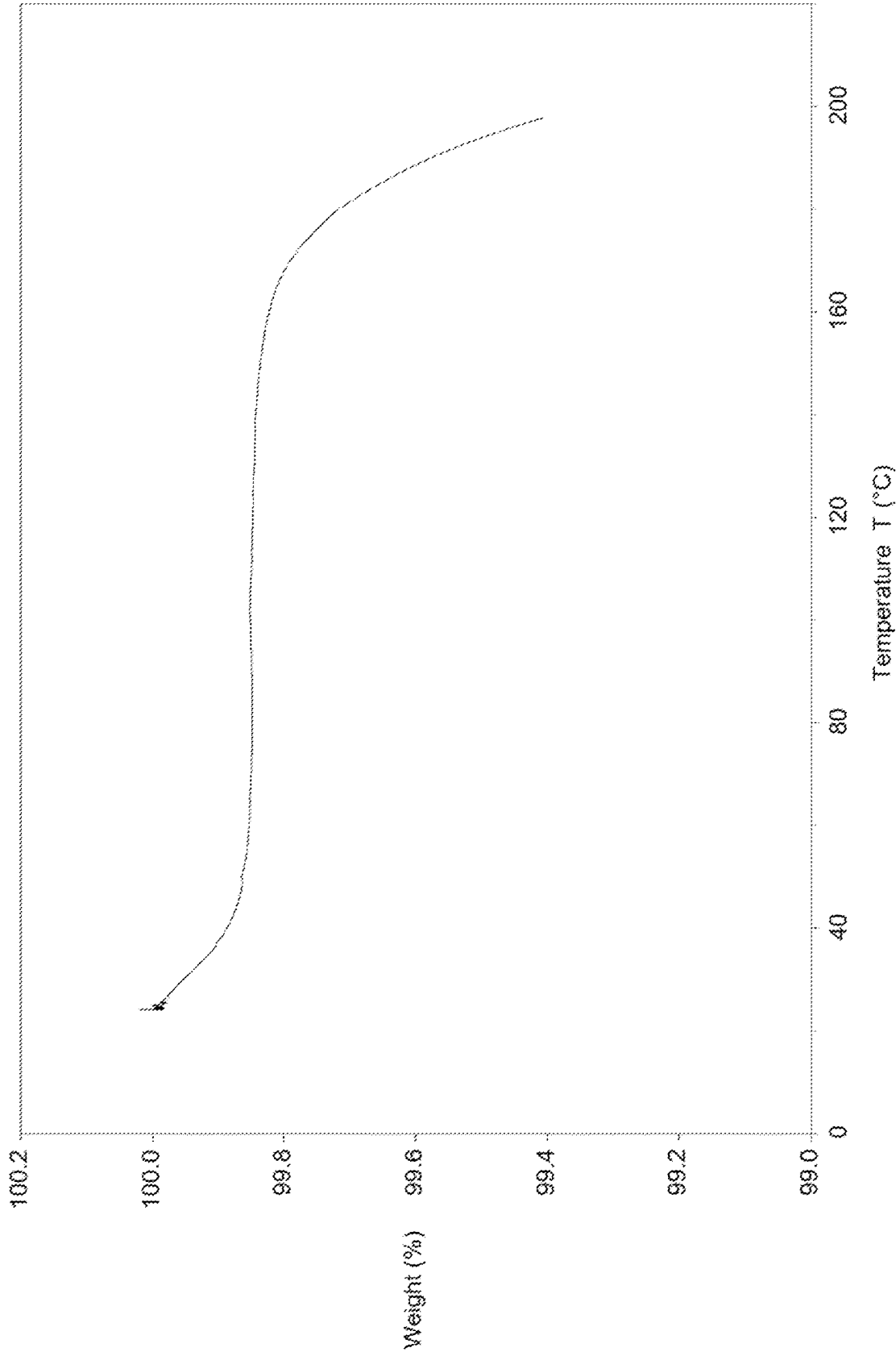


Fig. 8.

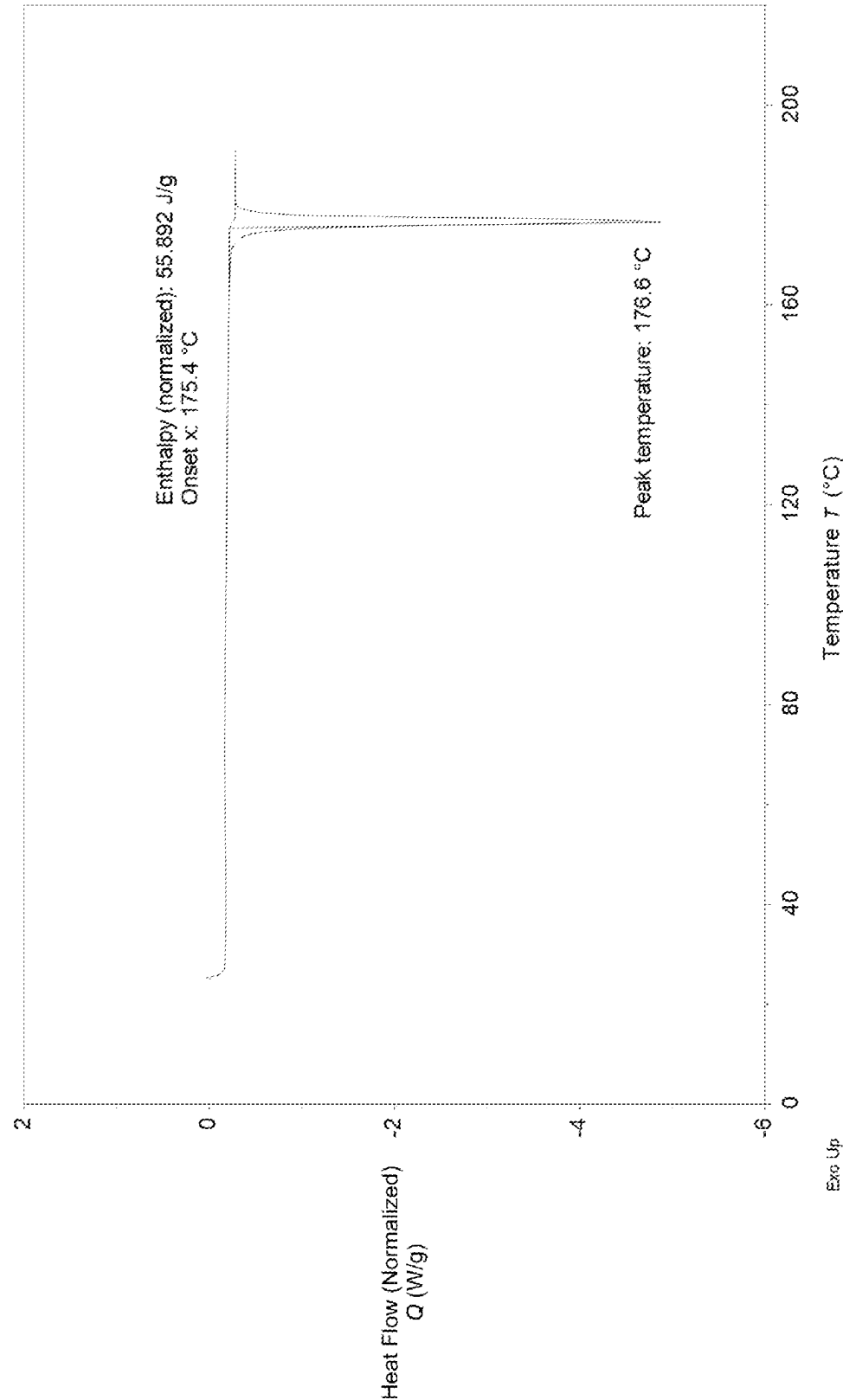


Fig. 9.

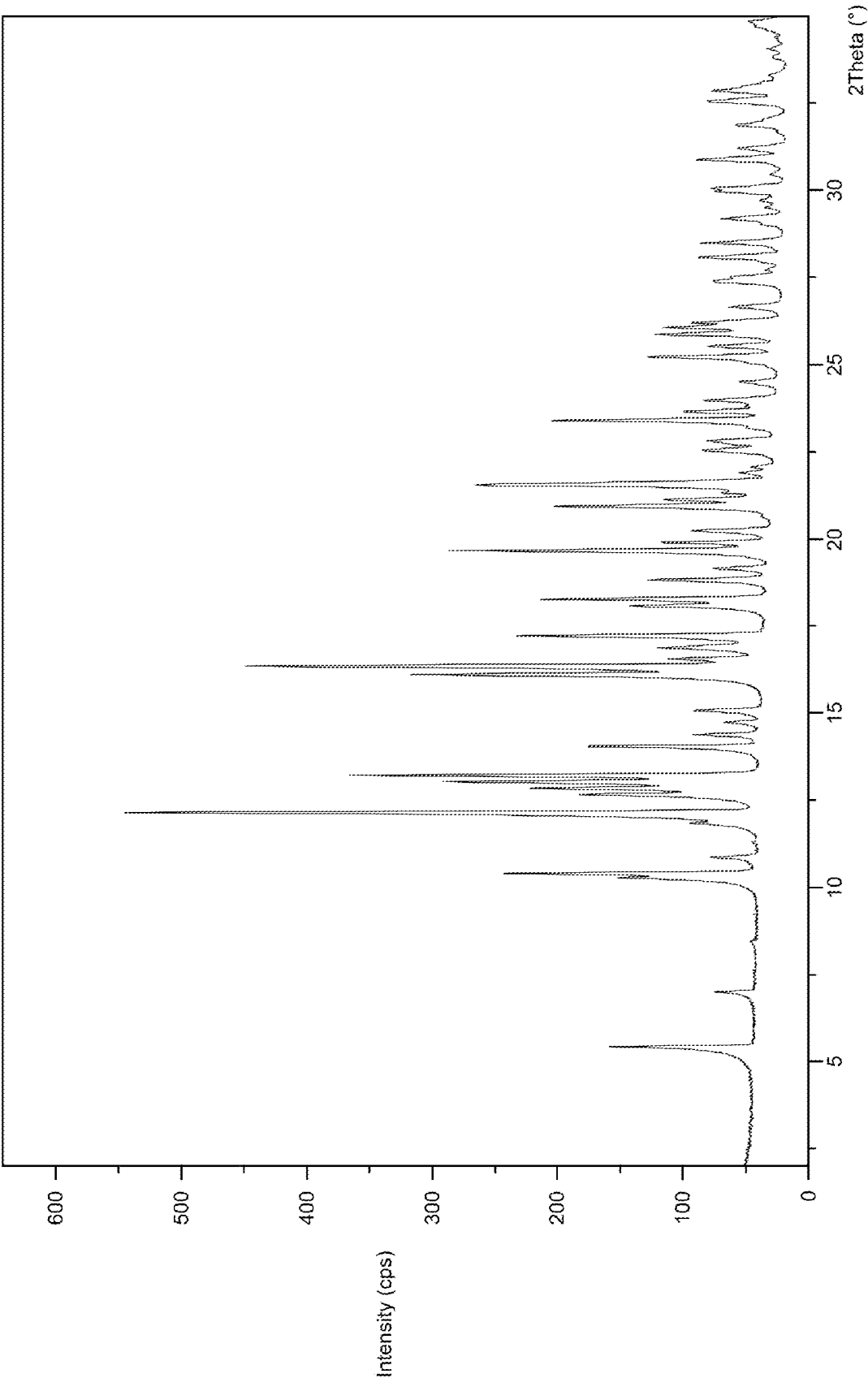


Fig. 10.

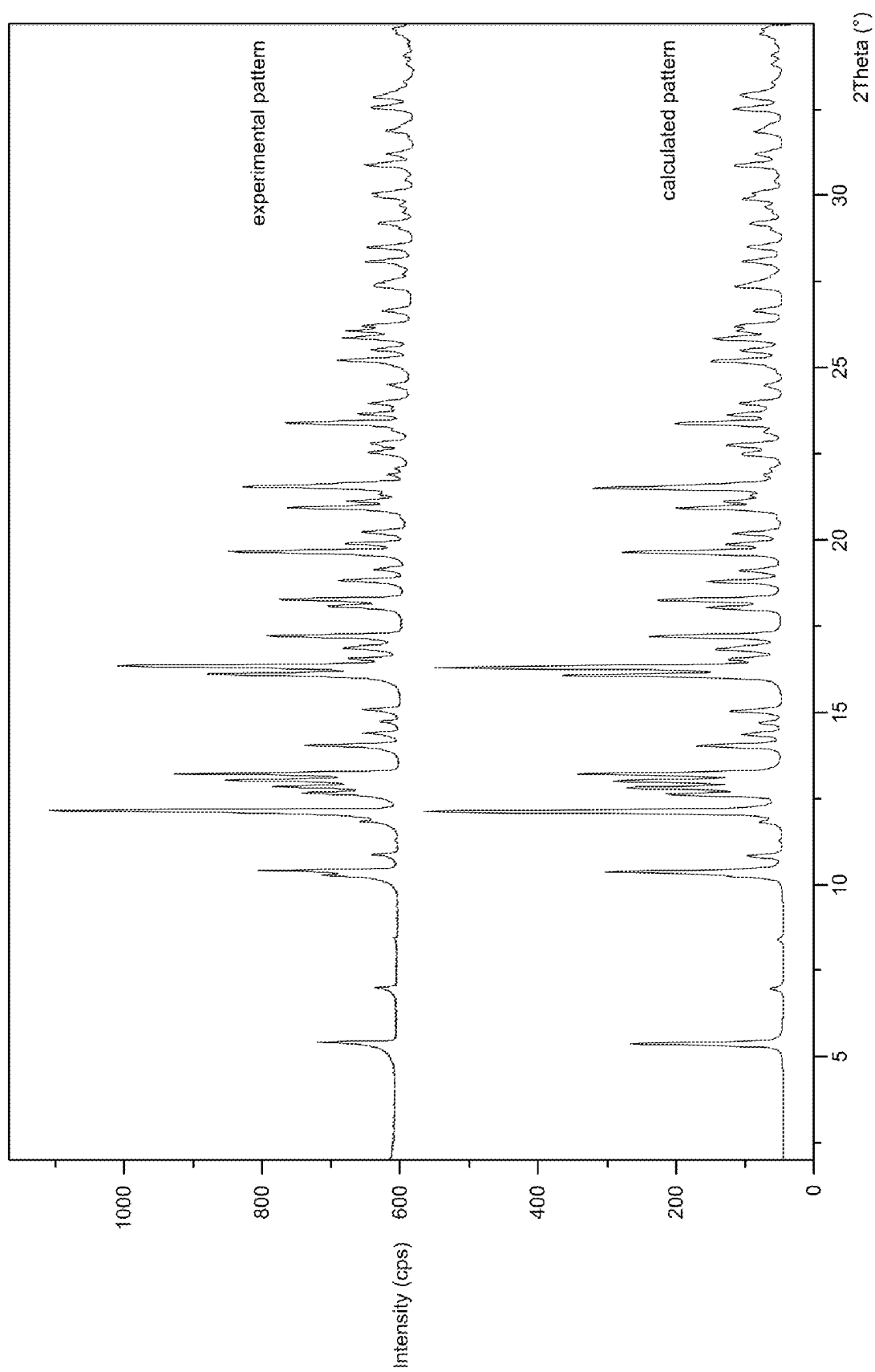


Fig. 11.

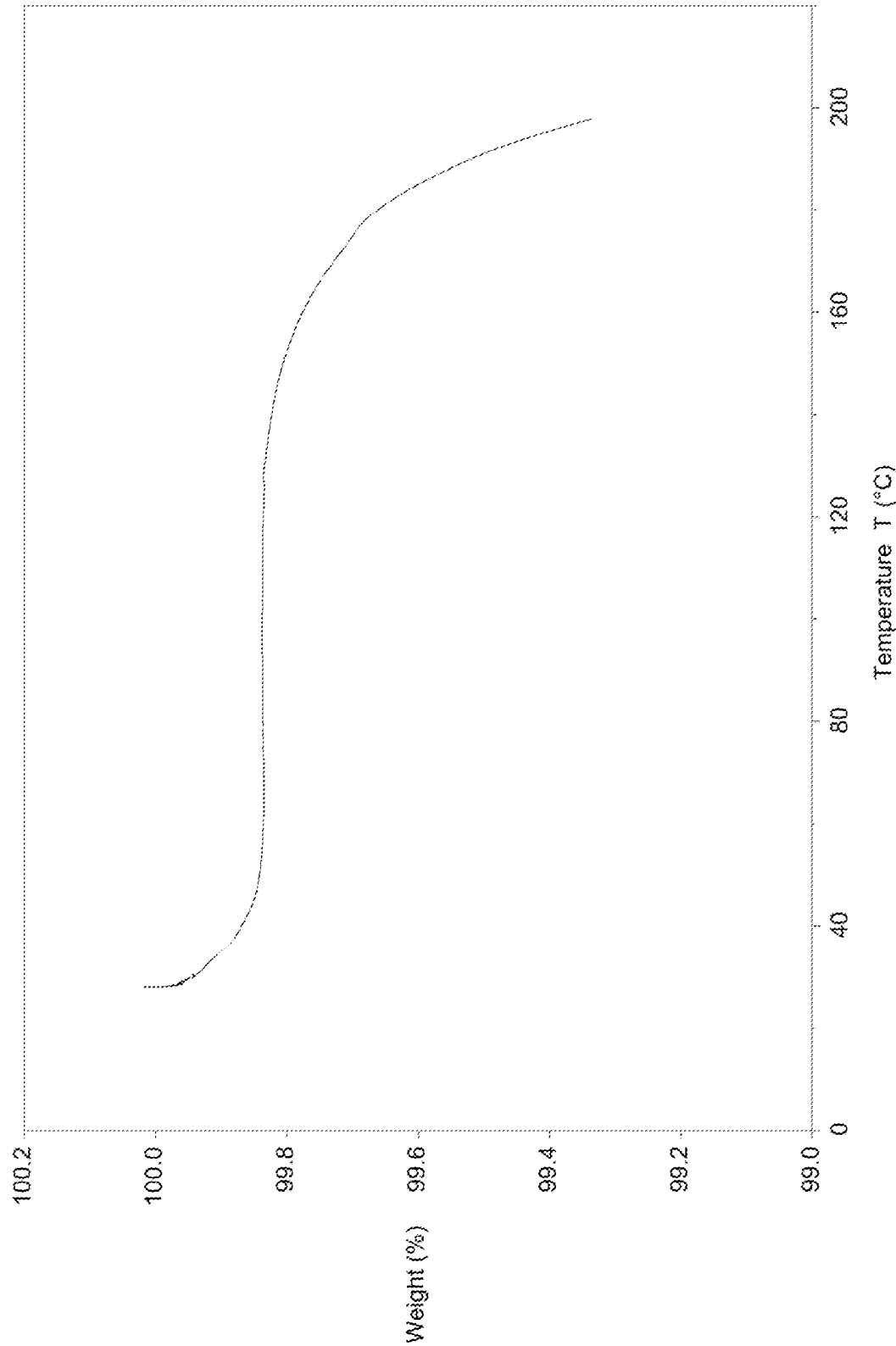


Fig. 12.

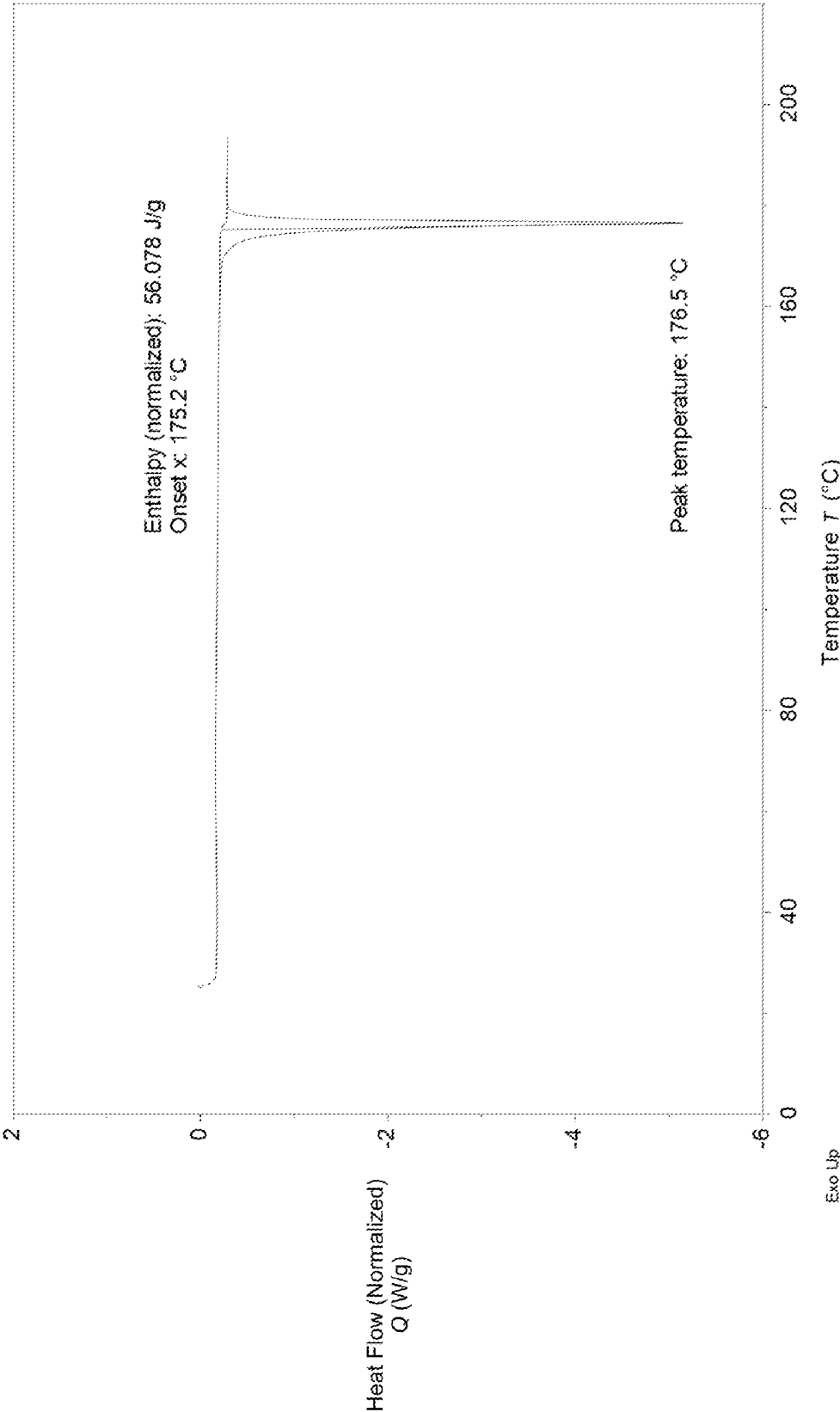


Fig. 13.

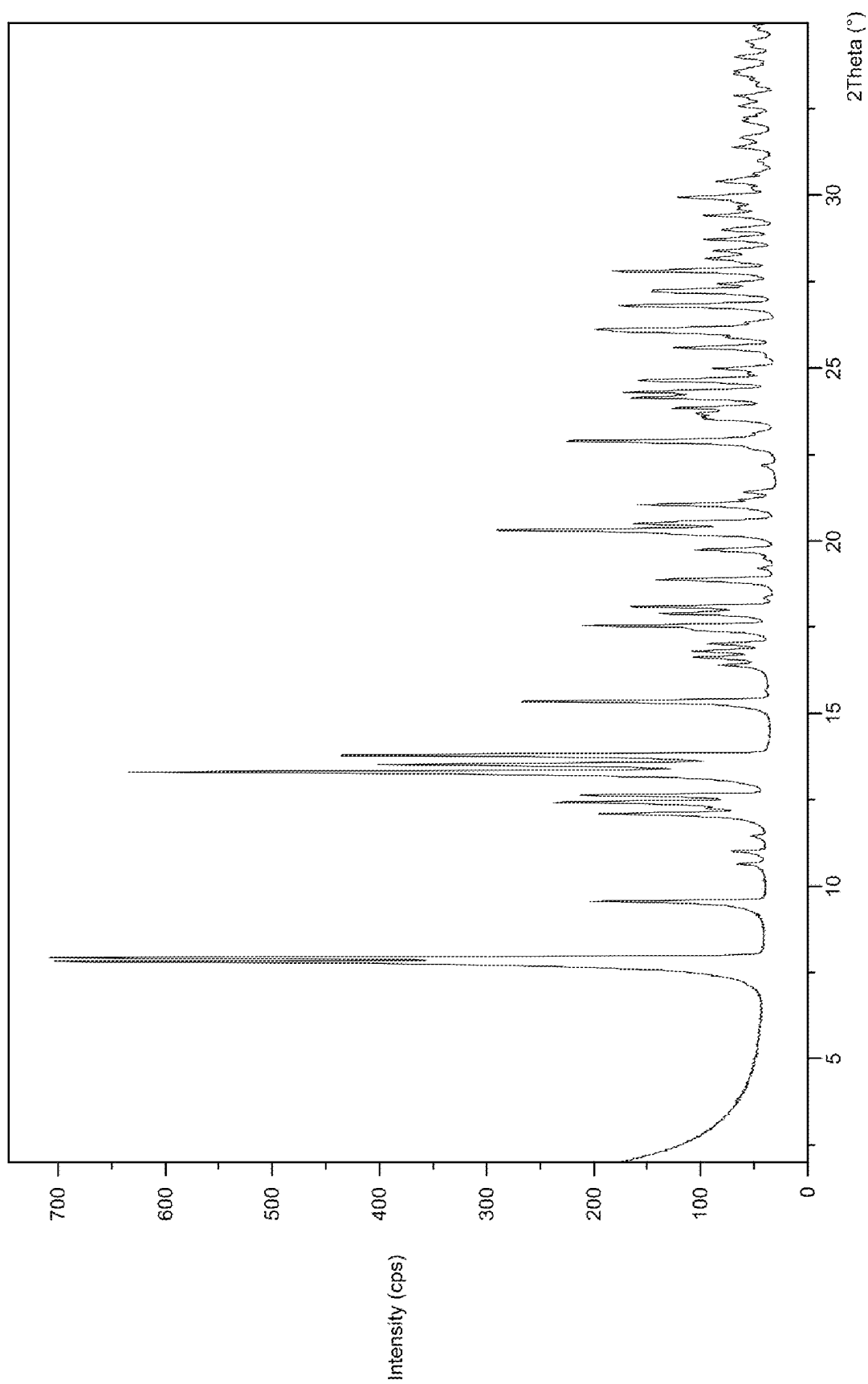
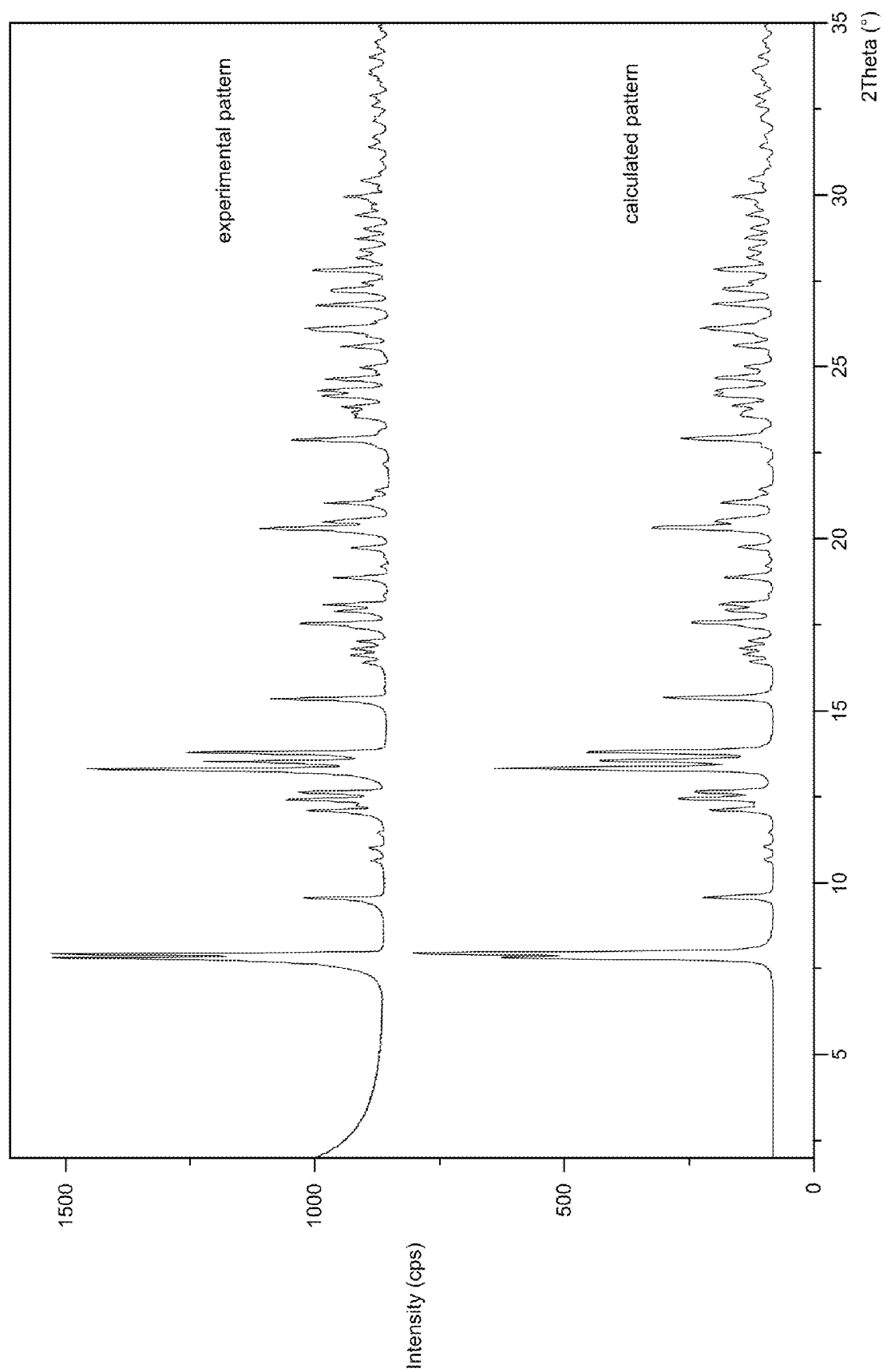


Fig. 14.



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BLARCAMESINE CO-CRYSTALS FOR THE MANUFACTURE OF PHARMACEUTICAL DOSAGE FORM

BACKGROUND OF THE INVENTION

Blarcamesine hydrochloride (ANAVEX2-73) is an investigational drug of Anavex Life Sciences that acts as a $\sigma 1$ receptor agonist and muscarinic receptor modulator. It is in Phase II-III human clinical trials and is being developed for the treatment of Alzheimer's disease and neuroprotective and neurodevelopmental disorders. It is also being developed for the treatment of Parkinson's disease-associated dementia (PDD) and rare paediatric diseases with orphan status (Rett syndrome and infantile spasms).

Blarcamesine molecule has a single asymmetric carbon unit.

The synthesis of racemic blarcamesine base is described in International Publications

WO 9730983 and WO 2013008044.

Several methods for the preparation of enantiomers of the blarcamesine base are described in International Publication WO 2013008044 and hydrochloric acid salt is also disclosed.

Three polymorphic forms of blarcamesine hydrochloride are disclosed in International Publication WO 2017013498 as well as a polymorphic form of a metabolite of blarcamesine base (ANAVEX19-144) prepared by two different methods. The metabolite is characterized in detail like blarcamesine HCl API.

International Publication WO 2019200345 of Anavex discloses the crystalline forms of blarcamesine in freebase and salt forms (hydrochloric acid salt and several other salts).

Teva Pharmaceuticals discloses some further salts in its International Publication

WO 2021158586.

SUMMARY OF THE INVENTION

The characteristics of solid forms of active pharmaceutical ingredient influence the key properties of the formulations they are used in, such as dissolution affecting bioavailability, stability and determine the range of applicable pharmaceutical technology procedures.

Therefore, the preparation of solid forms of active pharmaceutical ingredients (polymorphs, co-crystals, salts, solvates, complexes) and the study of their properties is an important and essential part of the drug development process.

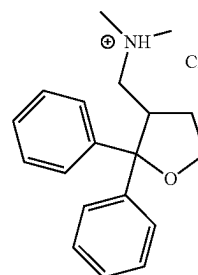
The object of our invention was to develop a new, pharmaceutically acceptable form of the active ingredient blarcamesine hydrochloride that would retain its uniformity during pharmaceutical processing (crushing, granulation, tableting), in stability studies required for drug development, and in use.

A further object was to develop a simple and economical process for the production of this form on an industrial scale.

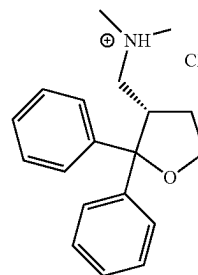
First aspect of our invention is the co-crystal of formula 2a consisting of $ZnCl_2$ and racemic-(±)-[(S)-(-), (R)-(+)] blarcamesine hydrochloride, as IUPAC name of [(2, 2-diphenyloxolan-3-yl)methyl]dimethylamine hydrochloride, of formula 1a; the co-crystal of formula 2b consisting of $ZnCl_2$ and (S)-(-)-blarcamesine hydrochloride of formula 1b and the co-crystal of formula 2c consisting of $ZnCl_2$ and (R)-(+)-blarcamesine hydrochloride of formula 1c.

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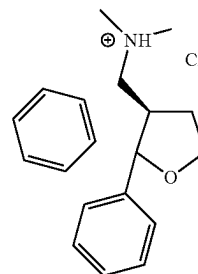
In the molecular formulas the dotted bonds relate to the coordinative bonds between the active pharmaceutical ingredient (blarcamesine hydrochloride) and co-crystal former ($ZnCl_2$; =“coformer”), in the same crystal lattice.



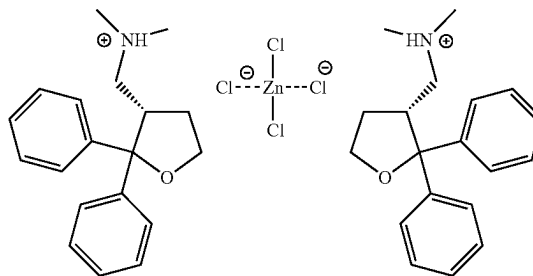
1a



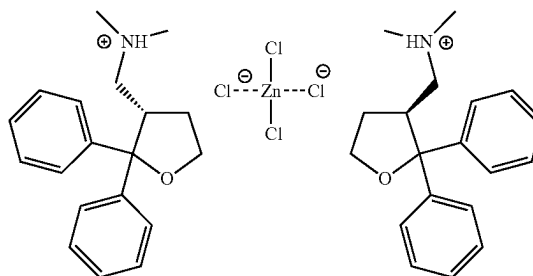
1b



1c



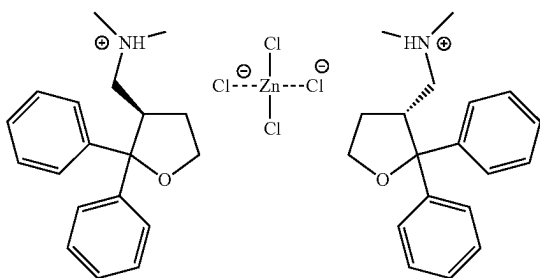
2a



2b

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-continued



Further aspect of our invention is the process for preparation of said co-crystals.

Further aspect of our invention is the pharmaceutical preparation comprising co-crystal according to our invention that comprises a therapeutically effective amount of co-crystal selected from formula 2a, 2b or 2c and one or more pharmaceutically acceptable excipients selected from filler, diluent, disintegrant, wetting agent, anti-adhesive agent, binder and other excipients commonly used in medicine.

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Further aspect of our invention is the process for preparation of said pharmaceutical preparation.

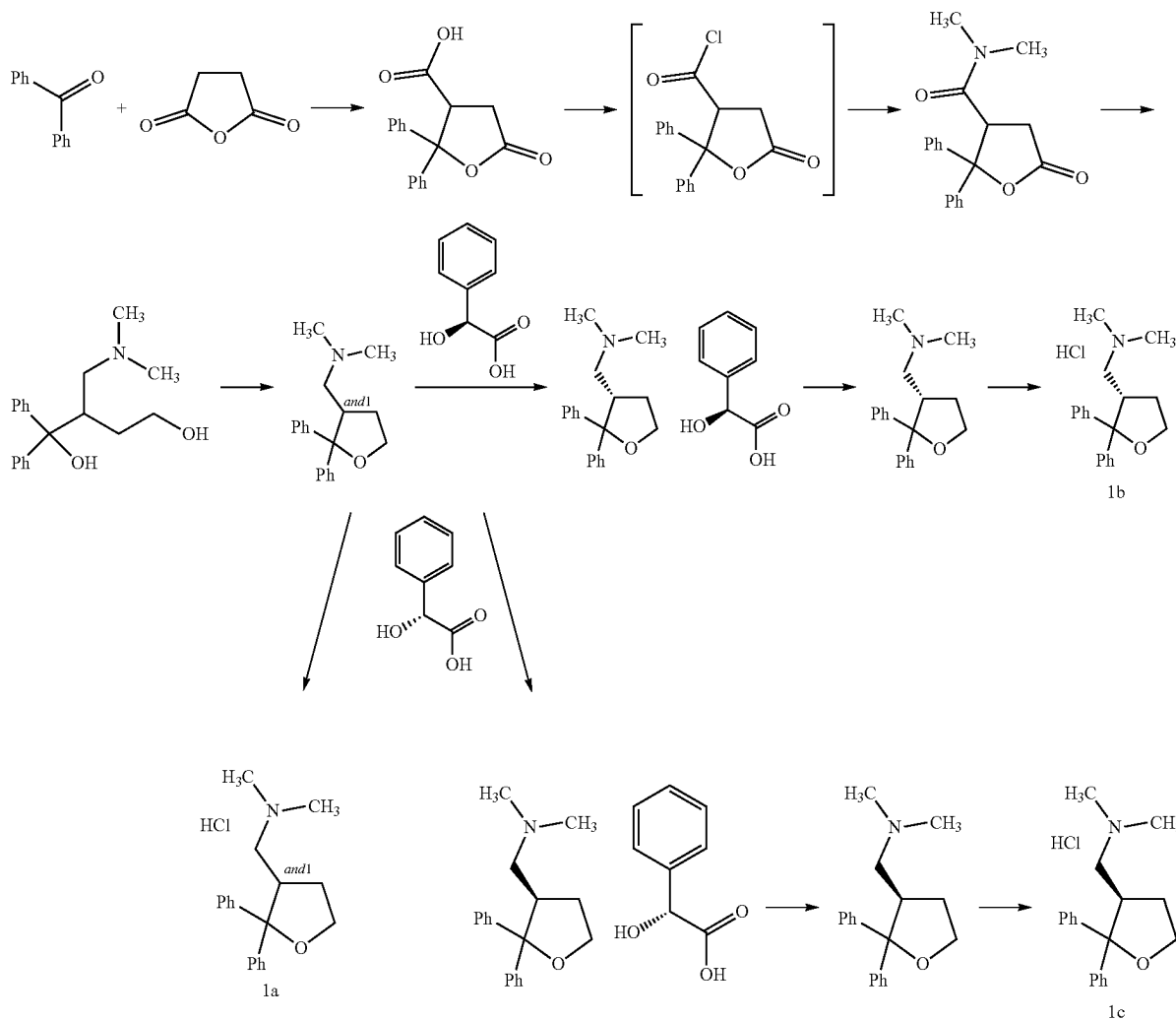
Further aspect of our invention is the use of co-crystal or a pharmaceutical composition containing co-crystal according to our invention in the treatment of neurodegenerative, neurodevelopmental diseases such as Alzheimer's disease, Parkinson's disease dementia and/or Rett syndrome.

Further aspect of our invention is the treatment of neurodegenerative, neurodevelopmental diseases such as Alzheimer's disease, Parkinson's disease dementia and/or Rett syndrome by administering a therapeutically effective amount of co-crystal or a medicinal product containing co-crystal according to our invention to the patient in need of treatment.

DETAILED DESCRIPTION OF THE INVENTION

The aim of our invention was achieved in the following way.

At first, we produced the racemic and optically active blarcamesine bases and their hydrochloric acid salts according to the process below, disclosed in the International Publication WO 9730983 and WO 2013008044 of Anavex:



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It is known that certain nitrogen-containing organic compounds having aromatic rings form well-crystallizing solid forms in the presence of inorganic salts.

Suitable inorganic salts can be found in several elements of the periodic table, such as the d group. In order to obtain a pharmaceutically acceptable form we had to choose an inorganic salt that is suitable for human use. Therefore Zn-, Mg-, Ca-compounds were investigated, and finally Zn, more precisely ZnCl_2 , was chosen.

A ZnCl_2 solution was added to the solution of racemic blarcamesine base with ether or alcohol and it was found that solid product was precipitated while standing.

Upon examining the solid product, we surprisingly found that the precipitated form was a co-crystal of the hydrochloric salt of blarcamesine with ZnCl_2 in a 2:1 molar ratio per unit cell.

The same solid form was also obtained from solutions of blarcamesine hydrochloride and ZnCl_2 .

We then focused on producing co-crystal with good reproducibility, robustness, and good chemical yield, and characterizing the resulting product.

As a result, we obtained the co-crystal by adding first ZnCl_2 and then hydrochloric acid solution to the mixture of the last chemical step of the process disclosed in the International Publication of WO 9730983 and WO 2013008044, without isolating of the racemic blarcamesine base, and finally filtering the product.

The process was carried out in organic solvent or mixtures thereof, which may be aprotic or protic, aromatic solvent, ether, or alcohol, preferably toluene, tetrahydrofuran (THF) or isopropyl alcohol (IPA), and mixtures thereof.

The temperature was between room temperature to 100° C., preferably between 70 and 80° C.

The properties of the isolated solid form were the same in all cases.

The chemical composition and physicochemical properties of the co-crystals were characterized by the following analytical methods: HPLC, ICP, $^1\text{H-NMR}$, $^{13}\text{C-NMR}$, COSY, HSQC, HMBC, elemental analysis, complexometric titration, SC-XRD, TG, DVS, DSC, XRPD.

Other properties of the of co-crystal were also investigated and compared with those of blarcamesine hydrochloride acid salt.

Surprisingly, we found that the ZnCl_2 co-crystal was more resistant to oxidative stress than the hydrochloric acid salt.

We also surprisingly found that the ZnCl_2 co-crystal retains its crystal structure after subjecting to wet grinding, whereas the polymorphic form of the hydrochloride salt changes under the same conditions.

As a further beneficial property, we found that the solubility of the ZnCl_2 co-crystal is excellent over the entire relevant pH range, but surprisingly much worse than the extreme solubility of the hydrochloric acid salt, which could be an advantageous feature both in the drug formulation technology process and in the potential implementation of a modified release formulation.

Based on the knowledge and experience with the racemic co-crystal, the corresponding optically active ZnCl_2 co-crystals were also successfully prepared from (S)-(-)- and (R)-(+)-blarcamesine bases, which were characterized in a similar way as described above.

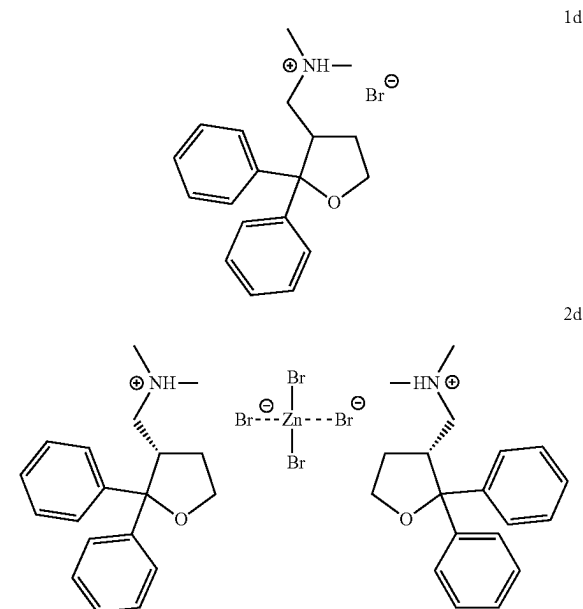
We carried out our experiments to prepare further co-crystals of the hydrogen halide salt of blarcamesine with Zn halide.

Therefore, we first prepared the racemic blarcamesine hydrobromide of formula 1d (as disclosed in International Publication of TEVA, WO 2021158586, Form A) and suc-

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cessfully prepared the racemic-(±)-blarcamesine hydrobromide ZnBr_2 (2:1 per unit cell) co-crystal of formula 2d in a similar way as described for the ZnCl_2 co-crystal.

The structure of the obtained solid form was confirmed by both powder and single-crystal X-ray diffraction studies.



BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Experimental X-ray powder diffractogram of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (1:1:1) co-crystal (2a)

FIG. 2: Experimental and calculated X-ray powder diffractograms of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (1:1:1) co-crystal (2a)

FIG. 3: TGA-thermogram of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (1:1:1) co-crystal (2a)

FIG. 4: DSC-thermogram of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (1:1:1) co-crystal (2a)

FIG. 5: Experimental X-ray powder diffractogram of (S)-(-)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2b)

FIG. 6: Experimental and calculated X-ray powder diffractograms of (S)-(-)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2b)

FIG. 7: TGA-thermogram of (S)-(-)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2b)

FIG. 8: DSC-thermogram of (S)-(-)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2b)

FIG. 9: Experimental X-ray powder diffractogram of (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2c)

FIG. 10: Experimental and calculated X-ray powder diffractograms of (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2c)

FIG. 11: TGA-thermogram of (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2c)

FIG. 12: DSC-thermogram of (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (2:1) co-crystal (2c)

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FIG. 13: Experimental X-ray powder diffractogram of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal (2d)

FIG. 14: Experimental and calculated X-ray powder diffractograms of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal (2d)

The single crystals prepared from the racemic-(±)-, (S)-(-)- and (R)-(+)-hydrochloride salts of structures 1a, 1b, 1c, 2a, 2b, and 2c respectively, furthermore the co-crystals formed from these salts with zinc chloride were measured with single-crystal X-ray diffractometry. Similarly, we investigated the single crystals of the racemic-(+)-blarcamesine hydrobromide salt of structure 1d and the co-crystal formed from this with zinc bromide. The results are summarized in Table 1. and Table 2.:

TABLE 1

Crystal parameters				
Compound	Space group	Cell constants	Density [kg/m ³]	N ⁺ ↔ Cl ⁻ distance [nm]
1a	P2 ₁ /a	a = 1.3527 nm b = 0.9182 nm c = 1.4247 nm β = 102.14°	1220	0.302
1b	P2 ₁ 2 ₁ 2 ₁	a = 0.7144 nm b = 1.4372 nm c = 1.7325 nm	1187	0.303
1c	P2 ₁ 2 ₁ 2 ₁	a = 0.7141 nm b = 1.4367 nm c = 1.7316 nm	1188	0.301
2a	P2 ₁ 2 ₁ 2 ₁	a = 1.5685 nm b = 1.5757 nm c = 1.6045 nm	1293	0.320
2b	P2 ₁ 2 ₁ 2 ₁	a = 0.8920 nm b = 1.3656 nm c = 3.2538 nm	1294	0.321
2c	P2 ₁ 2 ₁ 2 ₁	a = 0.8910 nm b = 1.3642 nm c = 3.2533 nm	1297	0.323

TABLE 2

Crystal parameters				
Compound	Space group	Cell constants	Density [kg/m ³]	N ⁺ ↔ Br ⁻ distance [nm]
1d	P2 ₁ /a	a = 1.4021 nm b = 0.9219 nm c = 1.4212 nm β = 102.34°	1341	0.316
2d	P2 ₁ 2 ₁ 2 ₁	a = 1.5438 nm b = 1.6018 nm c = 1.6581 nm	1539	0.337

We deduced two conclusions from the results measured by single-crystal X-ray diffraction, on one hand the co-crystals formed with zinc chloride differ from the starting hydrochloride salts without any doubt, on the other hand based on the N⁺ ↔ Cl⁻ distance the relation of the protonated ammonium group and the chloride ion did not change relevantly, which supports that really co-crystals were formed. Similarly, the co-crystal formed from the hydrobromide salt with zinc bromide differs without doubt from the starting hydrobromide salt and based on the N⁺ ↔ Br⁻ distance the relation between the protonated ammonium group and the bromide ion did not change relevantly, so in this case also a co-crystal was formed.

Characteristic X-ray powder diffraction peaks of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydro-

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chloride, ZnCl_2 (1:1:1 per unit cell) co-crystal (2a) are the following: 2θ (±0.2° 2θ): 7.93; 9.66; 17.82. More specifically it may be characterized by the following X-ray powder diffraction peaks: 2θ (±0.2° 2θ): 7.93; 9.66; 13.73; 15.77; 16.73; 17.82. Even more specifically it may be characterized by the following X-ray powder diffraction peaks: 2θ (±0.2° 2θ): 7.84; 7.93; 9.66; 11.25; 12.35; 12.49; 12.58; 13.58; 13.73; 15.77; 16.73; 16.84; 17.48; 17.82; 18.37; 18.63; 19.40; 20.04; 20.23; 20.32; 20.83; 20.96; 21.07; 22.11; 22.83; 23.04; 23.23; 23.91; 24.42; 24.88; 25.23; 25.51; 25.88; 26.32; 27.37; 27.69; 27.98; 28.10; 28.33; 28.54; 28.86; 29.41; 30.23; 30.47; 31.07; 32.33; 32.56; 32.86; 33.17; 33.55; 33.88; 34.10; 34.66. The characteristic X-ray powder diffractogram of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (1:1:1) co-crystal (2a) may be seen in FIG. 1, and the 2% or greater intensity peaks are summarized in Table 3.

TABLE 3

The X-ray powder diffraction data of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl_2 (1:1:1) co-crystal (relative intensities ≥ 2%)			
Peak No.	2θ (°)	d (Å)	Relative intensity (%)
1	7.84	11.27	26
2	7.93	11.15	43
3	9.66	9.15	11
4	11.25	7.86	3
5	12.35	7.17	16
6	12.49	7.09	19
7	12.58	7.04	31
8	13.58	6.52	100
9	13.73	6.45	52
10	15.77	5.62	17
11	16.73	5.30	13
12	16.84	5.26	2
13	17.48	5.07	19
14	17.82	4.98	56
15	18.37	4.83	17
16	18.63	4.76	8
17	19.40	4.58	3
18	20.04	4.43	7
19	20.23	4.39	13
20	20.32	4.37	19
21	20.83	4.26	20
22	20.96	4.24	32
23	21.07	4.22	11
24	22.11	4.02	9
25	22.83	3.89	4
26	23.04	3.86	3
27	23.23	3.83	15
28	23.91	3.72	19
29	24.42	3.64	9
30	24.88	3.58	10
31	25.23	3.53	13
32	25.51	3.49	10
33	25.88	3.44	16
34	26.32	3.39	4
35	27.37	3.26	12
36	27.69	3.22	7
37	27.98	3.19	5
38	28.10	3.18	7
39	28.33	3.15	3
40	28.54	3.13	14
41	28.86	3.09	21
42	29.41	3.04	6
43	30.23	2.96	5
44	30.47	2.93	6
45	31.07	2.88	7
46	32.33	2.77	2
47	32.56	2.75	9
48	32.86	2.73	2
49	33.17	2.70	4
50	33.55	2.67	5

TABLE 3-continued

The X-ray powder diffraction data of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl ₂ (1:1:1) co-crystal (relative intensities $\geq 2\%$)			
Peak No.	2 θ (°)	d (Å)	Relative intensity (%)
51	33.88	2.65	4
52	34.10	2.63	4
53	34.66	2.59	6

With single crystal X-ray diffraction, the exact atomic positions can be determined in the crystal. Based on these data, the powder X-ray diffraction pattern can be calculated. Experimental and calculated powder X-ray diffraction patterns of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (1:1:1) co-crystal (2a) are identical (FIG. 2), therefore the presented crystal phase is pure and uniform.

Based on the results of TGA (thermogravimetric analysis) measurements it can be stated that (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (1:1:1) co-crystal (2a) is a water and solvent free (anhydrous) form (amount of volatile compounds evaporated from the sample stayed below 0.1% w/w until reaching the temperature value of 160° C., FIG. 3) Based on DSC (differential scanning calorimetry) measurement the melting onset temperature of the presented crystal phase is 230-231° C. (FIG. 4)

Characteristic X-ray powder diffraction peaks of (S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1 per unit cell) co-crystal (2b) are the following: 2 θ ($\pm 0.2^\circ$ 2 θ): 12.15; 13.04; 19.67. More specifically it may be characterized by the following X-ray powder diffraction peaks: 2 θ ($\pm 0.2^\circ$ 2 θ): 12.15; 12.85; 13.04; 13.24; 16.36; 19.67. Even more specifically it may be characterized by the following X-ray powder diffraction peaks: 2 θ ($\pm 0.2^\circ$ 2 θ): 5.44; 7.02; 10.28; 10.41; 10.89; 11.82; 12.15; 12.67; 12.85; 13.04; 13.23; 14.05; 14.40; 14.75; 15.09; 16.11; 16.36; 16.56; 16.87; 17.23; 18.09; 18.28; 18.84; 19.17; 19.67; 19.90; 20.25; 20.94; 21.14; 21.34; 21.55; 21.90; 22.07; 22.56; 22.74; 22.85; 23.20; 23.41; 23.67; 23.98; 24.52; 25.22; 25.55; 25.88; 26.07; 26.22; 26.66; 27.39; 27.54; 27.75; 28.10; 28.51; 29.05; 29.20; 29.54; 29.74; 30.00; 30.09; 30.48; 30.90; 31.23; 31.90; 32.04; 32.58; 32.86; 33.32; 33.81; 34.08; 34.71; 34.88. The characteristic X-ray powder diffractogram of (S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal (2b) may be seen in FIG. 5, and the 2% or greater intensity peaks are summarized in Table 4.

TABLE 4

The X-ray powder diffraction data of (S)-(-)-blarcamesine hydrochloride, ZnCl ₂ (2:1) co-crystal (relative intensities $\geq 2\%$)			
Peak No.	2 θ (°)	d (Å)	Relative intensity (%)
1	5.44	16.24	17
2	7.02	12.60	6
3	10.28	8.60	24
4	10.41	8.49	20
5	10.89	8.12	6
6	11.82	7.49	15
7	12.15	7.29	100
8	12.67	6.98	19
9	12.85	6.89	26
10	13.04	6.79	41
11	13.23	6.69	62

TABLE 4-continued

The X-ray powder diffraction data of (S)-(-)-blarcamesine hydrochloride, ZnCl ₂ (2:1) co-crystal (relative intensities $\geq 2\%$)			
Peak No.	2 θ (°)	d (Å)	Relative intensity (%)
12	14.05	6.30	24
13	14.40	6.15	9
14	14.75	6.01	5
15	15.09	5.87	8
16	16.11	5.50	40
17	16.36	5.42	60
18	16.56	5.35	16
19	16.87	5.25	17
20	17.23	5.15	39
21	18.09	4.90	16
22	18.28	4.85	32
23	18.84	4.71	15
24	19.17	4.63	5
25	19.67	4.51	46
26	19.90	4.46	23
27	20.25	4.39	10
28	20.94	4.24	42
29	21.14	4.20	16
30	21.34	4.16	4
31	21.55	4.12	56
32	21.90	4.06	4
33	22.07	4.03	3
34	22.56	3.94	13
35	22.74	3.91	5
36	22.85	3.89	5
37	23.20	3.83	2
38	23.41	3.80	34
39	23.67	3.76	16
40	23.98	3.71	15
41	24.52	3.63	6
42	25.22	3.53	18
43	25.55	3.49	9
44	25.88	3.44	18
45	26.07	3.42	16
46	26.22	3.40	16
47	26.66	3.34	10
48	27.39	3.26	12
49	27.54	3.24	5
50	27.75	3.21	2
51	28.10	3.18	14
52	28.51	3.13	16
53	29.05	3.07	3
54	29.20	3.06	11
55	29.54	3.02	3
56	29.74	3.00	2
57	30.00	2.98	7
58	30.09	2.97	10
59	30.48	2.93	2
60	30.90	2.89	21
61	31.23	2.86	10
62	31.90	2.81	10
63	32.04	2.79	3
64	32.58	2.75	13
65	32.86	2.73	22
66	33.32	2.69	4
67	33.81	2.65	3
68	34.08	2.63	3
69	34.71	2.58	6
70	34.88	2.57	9

With single crystal X-ray diffraction, the exact atomic positions can be determined in the crystal. Based on these data, the powder X-ray diffraction pattern can be calculated. Experimental and calculated powder X-ray diffraction patterns of (S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal (2b) are identical (FIG. 6), therefore the presented crystal phase is pure and uniform.

Based on the results of TGA (thermogravimetric analysis) measurements it can be stated that (S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal (2b) is a water and solvent free (anhydrous) form (amount of volatile compounds evaporated from the sample stayed below 0.5% w/w

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until reaching the temperature value of 120° C., FIG. 7) Based on DSC (differential scanning calorimetry) measurement the melting onset temperature of the presented crystal phase is 175-176° C. (FIG. 8)

Characteristic X-ray powder diffraction peaks of (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1 per unit cell) co-crystal (2c) are the following: 2θ (±0.2° 2θ): 12.15; 13.04; 19.67. More specifically it may be characterized by the following X-ray powder diffraction peaks: 2θ (±0.2° 2θ): 12.15; 12.85; 13.04; 13.24; 16.36; 19.67. Even more specifically it may be characterized by the following X-ray powder diffraction peaks: 2θ (±0.2° 2θ): 5.43; 7.01; 10.27; 10.40; 10.87; 11.82; 12.15; 12.65; 12.85; 13.04; 13.24; 14.04; 14.38; 14.73; 15.07; 16.09; 16.36; 16.54; 16.85; 17.21; 18.07; 18.27; 18.81; 19.14; 19.67; 19.88; 20.23; 20.93; 21.13; 21.31; 21.52; 21.88; 22.06; 22.53; 22.72; 22.81; 23.17; 23.39; 23.65; 23.97; 24.50; 25.21; 25.52; 25.86; 26.05; 26.20; 26.64; 27.37; 27.52; 27.71; 28.07; 28.48; 29.02; 29.18; 29.52; 29.71; 29.96; 30.07; 30.45; 30.88; 31.20; 31.87; 32.04; 32.54; 32.84; 33.30; 33.79; 34.05; 34.69; 34.85. The characteristic X-ray powder diffractogram of (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal (2c) may be seen in FIG. 9, and the 2% or greater intensity peaks are summarized in Table 5. (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal and (S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal have the same X-ray powder diffraction pattern.

TABLE 5

The X-ray powder diffraction data of (R)-(+)-blarcamesine hydrochloride, ZnCl ₂ (2:1) co-crystal (relative intensities ≥ 2%)			
Peak No.	2θ (°)	d (Å)	Relative intensity (%)
1	5.43	16.29	29
2	7.01	12.62	7
3	10.27	8.62	21
4	10.40	8.51	39
5	10.87	8.14	10
6	11.82	7.49	11
7	12.15	7.29	100
8	12.65	7.00	26
9	12.84	6.90	33
10	13.04	6.80	48
11	13.24	6.70	64
12	14.04	6.31	30
13	14.38	6.16	12
14	14.73	6.01	7
15	15.07	5.88	14
16	16.09	5.51	55
17	16.34	5.43	93
18	16.54	5.36	17
19	16.85	5.26	24
20	17.21	5.15	42
21	18.07	4.91	22
22	18.27	4.86	36
23	18.81	4.72	20
24	19.14	4.64	9
25	19.67	4.52	50
26	19.88	4.47	12
27	20.23	4.39	15
28	20.93	4.24	38
29	21.13	4.21	16
30	21.31	4.17	7
31	21.52	4.13	70
32	21.88	4.06	3
33	22.06	4.03	2
34	22.53	3.95	14
35	22.72	3.91	6
36	22.81	3.90	9
37	23.17	3.84	3
38	23.39	3.80	36
39	23.65	3.76	15

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TABLE 5-continued

The X-ray powder diffraction data of (R)-(+)-blarcamesine hydrochloride, ZnCl ₂ (2:1) co-crystal (relative intensities ≥ 2%)			
Peak No.	2θ (°)	d (Å)	Relative intensity (%)
40	23.97	3.71	20
41	24.50	3.63	6
42	25.21	3.53	21
43	25.52	3.49	12
44	25.86	3.45	22
45	26.05	3.42	16
46	26.20	3.40	17
47	26.64	3.35	10
48	27.37	3.26	14
49	27.52	3.24	6
50	27.71	3.22	2
51	28.07	3.18	14
52	28.48	3.13	15
53	29.02	3.08	5
54	29.18	3.06	10
55	29.52	3.03	3
56	29.71	3.01	4
57	29.96	2.98	11
58	30.07	2.97	11
59	30.45	2.94	2
60	30.88	2.90	19
61	31.20	2.87	11
62	31.87	2.81	11
63	32.04	2.79	4
64	32.54	2.75	18
65	32.84	2.73	23
66	33.30	2.69	4
67	33.79	2.65	4
68	34.05	2.63	4
69	34.69	2.59	6
70	34.85	2.57	11

With single-crystal X-ray diffraction the exact atomic positions can be determined in the crystal. Based on these data, the powder X-ray diffraction pattern can be calculated. Experimental and calculated powder X-ray diffraction patterns of (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal (2c) are identical (FIG. 10), therefore the presented crystal phase is pure and uniform.

Based on the results of TGA (thermogravimetric analysis) measurements it can be stated that (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal (2c) is a water and solvent free (anhydrous) form (amount of volatile compounds evaporated from the sample stayed below 0.5% w/w until reaching the temperature value of 120° C., FIG. 11) Based on DSC (differential scanning calorimetry) measurement the melting onset temperature of the presented crystal phase is 175-176° C. (FIG. 12)

Characteristic X-ray powder diffraction peaks of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr₂ (1:1:1 per unit cell) co-crystal (2d) are the following: 2θ (±0.2° 2θ): 7.81; 9.57; 13.30. More specifically it may be characterized by the following X-ray powder diffraction peaks: 2θ (±0.2° 2θ): 7.81; 9.57; 13.30; 15.35; 24.30. Even more specifically it may be characterized by the following X-ray powder diffraction peaks: 2θ (±0.2° 2θ): 7.81; 7.93; 9.57; 10.67; 11.02; 11.44; 12.10; 12.23; 12.42; 12.63; 13.30; 13.52; 13.79; 15.35; 16.40; 16.61; 16.79; 17.01; 17.40; 17.53; 17.89; 18.08; 18.86; 19.73; 20.17; 20.30; 20.48; 21.04; 21.18; 21.40; 22.68; 22.88; 23.11; 23.53; 23.68; 23.84; 24.14; 24.30; 24.64; 24.82; 24.98; 25.58; 25.88; 26.04; 26.11; 26.31; 26.80; 27.20; 27.43; 27.81; 28.05; 28.17; 28.39; 28.72; 29.00; 29.41; 29.60; 29.81; 29.94; 30.15; 30.41; 30.64; 31.01; 31.39; 31.65; 32.14; 32.26; 32.58; 32.89; 33.17; 33.35; 33.51; 33.62; 34.00; 34.28; 34.45; 34.78; 34.89. The characteristic X-ray

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powder diffractogram of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal (2d) may be seen in FIG. 13, and the 2% or greater intensity peaks are summarized in Table 6.

TABLE 6

The X-ray powder diffraction data of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal (relative intensities $\geq 2\%$)			
Peak No.	2 θ (°)	d (Å)	Relative intensity (%)
1	7.81	11.32	100
2	7.93	11.15	66
3	9.57	9.25	25
4	10.67	8.29	10
5	11.02	8.03	6
6	11.44	7.73	4
7	12.10	7.32	23
8	12.23	7.23	5
9	12.42	7.13	23
10	12.63	7.01	21
11	13.30	6.66	74
12	13.52	6.55	44
13	13.79	6.42	52
14	15.35	5.77	31
15	16.40	5.41	6
16	16.61	5.34	9
17	16.79	5.28	9
18	17.01	5.21	7
19	17.40	5.10	8
20	17.53	5.06	20
21	17.89	4.96	12
22	18.08	4.91	16
23	18.86	4.70	13
24	19.73	4.50	8
25	20.17	4.40	7
26	20.30	4.38	31
27	20.48	4.34	18
28	21.04	4.22	17
29	21.18	4.19	3
30	21.40	4.15	3
31	22.68	3.92	3
32	22.88	3.89	26
33	23.11	3.85	2
34	23.53	3.78	9
35	23.68	3.76	7
36	23.84	3.73	12
37	24.14	3.69	15
38	24.30	3.66	20
39	24.64	3.61	18
40	24.82	3.59	2
41	24.98	3.57	7
42	25.58	3.48	12
43	25.88	3.44	3
44	26.04	3.42	12
45	26.11	3.41	22
46	26.31	3.39	3
47	26.80	3.33	19
48	27.20	3.28	22
49	27.43	3.25	5
50	27.81	3.21	21
51	28.05	3.18	4
52	28.17	3.17	10
53	28.39	3.14	8
54	28.72	3.11	9
55	29.00	3.08	8
56	29.41	3.04	11
57	29.60	3.02	5
58	29.81	3.00	5
59	29.94	2.98	14
60	30.15	2.96	5
61	30.41	2.94	10
62	30.64	2.92	4
63	31.01	2.88	4
64	31.39	2.85	9
65	31.65	2.83	9
66	32.14	2.78	6
67	32.26	2.77	8

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TABLE 6-continued

The X-ray powder diffraction data of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal (relative intensities $\geq 2\%$)			
Peak No.	2 θ (°)	d (Å)	Relative intensity (%)
68	32.58	2.75	8
69	32.89	2.72	7
70	33.17	2.70	5
71	33.35	2.69	6
72	33.51	2.67	7
73	33.62	2.67	5
74	34.00	2.64	9
75	34.28	2.62	5
76	34.45	2.60	8
77	34.78	2.58	4
78	34.89	2.57	5

Measurement conditions	
Single-crystal X-ray diffraction parameters	
Instrument:	Rigaku R-Axis SPIDER diffractometer
Detection:	Image plate X-ray tube
Type:	Rigaku Long Fine Focus
Anode:	Cu
Wavelength:	K α (1.541874 Å)
X-ray powder diffraction instrument settings	
Instrument:	PANalytical Empyrean X-ray powder diffractometer
Sample mode:	Transmission X-ray tube
Type:	Empyrean Long Fine Focus High Resolution tube
Anode material:	Cu
Wavelength:	K α (1.541874 Å)
Focus mode:	line focus
Incident beam optics	
Divergence slit:	Fixed slit 1/2°
Mirror:	Focusing elliptical mirror
Soller slit:	0.04 rad
Anti-scatter slit:	Fixed slit 1/2°
Diffracted beam optics	
Anti-scatter slit:	Programmable slit in fix mode: 1/2°
Soller slit:	0.04 rad
Sample stage	
Type:	Reflection-transmission spinner stage
Sample rotation:	1 rps
Beam knife:	Transmission beam stop used
Detector	
Type:	PIXcel 3D 1 × 1 area detector
Mode:	Scanning line detector (1D) mode
Active length:	3.3473°
Sample preparation:	place powder samples (without grinding) between two Mylar foils in the sample holder

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-continued

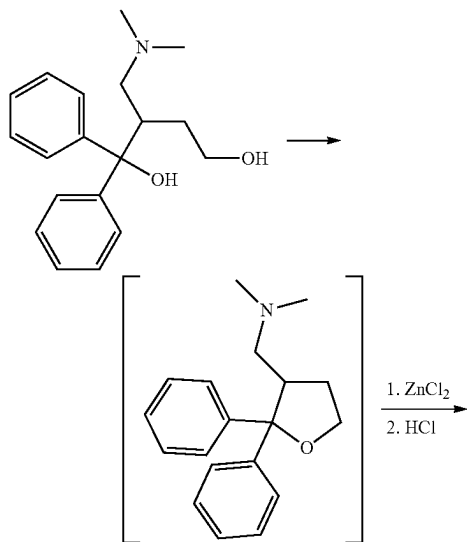
Measurement conditions	
Measurement settings	
Temperature:	room temperature
Accelerating voltage:	45 kV
Anode current:	40 mA
Scan type:	continuous gonio (θ/θ) scan
Measurement range:	range: 2.0000 – 34.9964 $^{\circ}2\theta$
Step size:	0.0131 $^{\circ}2\theta$
Time per step:	109.650 s
Measurement cycles:	1
Measurement time:	~20 minutes
Thermogravimetry instrument settings	
Device:	TA Instruments Discovery TGA thermogravimetric analyzer
Atmosphere:	N ₂ flow: 25 mL/min (furnace) 10 mL/min (balance)
Data sampling interval:	0.5 s/pt
Temperature program (2a):	30° C.-240° C. 10° C./min
Temperature program (2b, 2c):	30° C.-200° C. 10° C./min
Pan:	Platinum 100 μ L
Differential scanning calorimetry instrument settings	
Device:	TA Instruments Discovery DSC differential scanning calorimeter
Atmosphere:	N ₂ flow (50 mL/min)
Data sampling interval:	0.1 s/pt
Temperature program (2a):	35° C.-250° C. 10° C./min
Temperature program (2b, 2c):	30° C.-190° C. 10° C./min
Pan (2a)	Hermetically sealed Al
Pan (2b, 2c)	Standard Al sealed

EXAMPLES

The following examples are included for the purpose of illustration of the disclosure and are to be construed by way of example and not as a limitation.

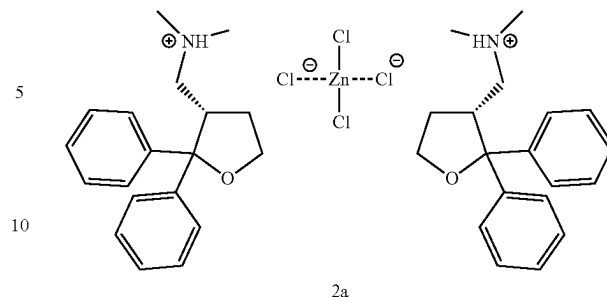
Example 1

Preparation of (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (1:1:1) co-crystal (2a)



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-continued



To a round bottom flask filled with toluene solution of 2-[(dimethylamino)methyl]-1,1-diphenylbutane-1,4-diol (3.68 g, 12.29 mmol) p-toluenesulfonic acid was added. The mixture was refluxed with Dean-Stark apparatus for water removal. After the completion of the reaction the toluene solution were washed with aq. NaOH solution then brine, and dried. Isopropyl alcohol (IPA) (12 mL) was added to the residual 40 ml solution and warmed to 80° C. While continuous stirring, at first a pre-prepared 80° C. toluene-IPA (1:1) ZnCl₂ solution (0.92 g, 6.76 mmol, 0.55 mol eq., dissolved in 6 mL solvent) then a 6.25 M IPA/HCl solution (2.16 mL, 13.52 mmol, 1.1 mol eq.) was added. The mixture was stirred further, and the precipitated solid was filtered off still in warm. The solid crystalline material was washed and dried in vacuum until constant weight, obtaining 4.34 g (91.5%) white (S)-(-)-blarcamesine hydrochloride, (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (1:1:1) co-crystal.

HPLC purity: 99.5%.

HPLC purity: 99.97% (after crystallization from aq. IPA)

Mp.: 234-236° C.

Zn content (complexometric titration): 8.67% (calculated: 8.47%)

¹H-NMR (DMSO, 600 MHz): 9.94 (bs, 1H); 7.63 (~d, J=8.4 Hz, 2H); 7.42 (~d, J=8.3 Hz, 2H); 7.35 (m, 2H); 7.31 (m, 2H); 7.24 (~t, J=7.3 Hz, 1H); 7.20 (~t, J=7.3 Hz, 1H); 4.17 (q, J=7.8 Hz 1H); 3.76 (td, J=8.8, 4.7 Hz, 1H); 3.59 (m, 1H); 2.83 (bs, 3H); 2.73 (bs, 3H); 2.72 (m, 2H); 2.25 (m, 1H); 1.92 (m, 1H).

¹³C-NMR (DMSO, 150 MHz): 145.51; 142.64; 128.62; 128.50; 127.24; 127.01; 126.01; 125.57; 89.32; 64.58; 57.89; 44.86; 41.33; 40.99; 28.52.

COSY: 9.94-2.83; 2.73; 7.63-7.35-7.24; 7.42-7.31-7.20; (4.17; 3.76)-(2.25; 1.92)-3.59; 2.73.

HSQC: 7.63-125.57; 7.42-126.01; 7.35-128.62; 7.31-128.50; 7.24-127.24; 7.20-127.01; (4.17; 3.76)-64.58; 3.59-40.99; 2.83-41.33; 2.73-44.86; 2.72-57.89; (2.25; 1.92)-28.52.

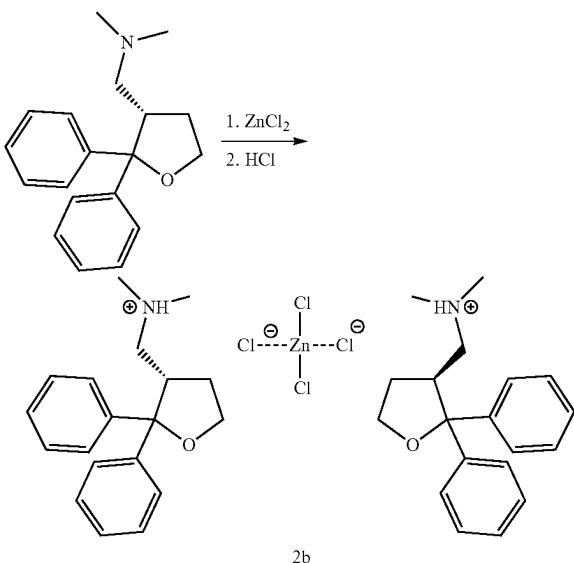
HMBC (characteristic cross peaks): 7.63-89.32; 7.42-89.32; 7.35-145.51; 7.31-142.64; 4.17-89.32.

Elemental analysis C₃₈H₄₈Cl₄N₂O₂Zn (M: 771.99) calculated: C, 59.12%; H, 6.27%; N, 3.63%; Cl, 18.37% measured: C, 59.17%; H, 6.39%; N, 3.57%; Cl, 18.46%

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Example 2

(S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1)
co-crystal (2b)



(S)-(-)-blarcamesine base (1.50 g, 5.33 mmol) was dissolved in IPA (10 mL), then warmed to 80° C. A pre-prepared 80° C. IPA solution of ZnCl₂ (0.40 g, 2.93 mmol, dissolved in 5 mL solvent) then a 6.25 M IPA/HCl solution (0.94 mL, 5.86 mmol, 1.1 mol eq.) was added. The mixture was stirred further, and the precipitated solid was filtered off still in warm. The solid crystalline material was dried in vacuum until constant weight, yielding 1.80 g (87.5%) white (S)-(-)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal.

HPLC purity: 99.9%

Mp.: 176-178° C.

Zn content (complexometric titration): 8.76% (calculated: 8.47%)

¹H-NMR (DMSO, 600 MHz): 9.90 (b, 1H); 7.62 (m, 2H); 7.41 (m, 2H); 7.35 (m, 2H); 7.31 (m, 2H); 7.24 (m, 1H); 7.20 (m, 1H); 4.17 (m, 1H); 3.76 (m, 1H); 3.58 (m, 1H); 2.75 (b, 3H); 2.75 (b, 3H); 2.70 (b, 2H); 2.24 (m, 1H); 1.92 (m, 1H).

¹³C-NMR (DMSO, 150 MHz): 145.53; 142.66; 128.63; 128.50; 127.25; 127.01; 126.02; 125.58; 89.31; 64.59; 57.95; 44.85.

COSY: 7.62-7.35-7.24; 7.41-7.31-7.20; (4.17; 3.76)-(2.24; 1.92)-3.58; 2.70.

HSQC (140 Hz): 7.62-125.58; 7.41-126.02; 7.5-128.63; 7.31-128.50; 7.24-127.25; 7.20-127.01; 4.17-64.59; 3.76-64.59; 3.58-41.04; 2.75-41.37; 2.75-44.85; 2.70-57.95; 2.24-28.52; 1.92-28.52.

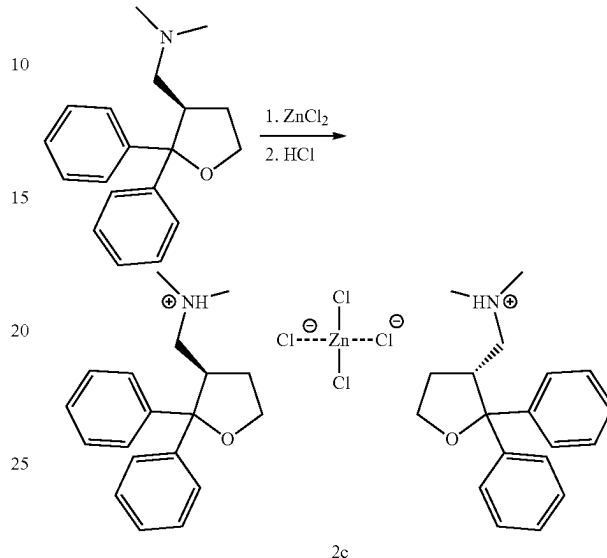
HMBC (8 Hz, 140 Hz): 7.62-(127.25; 125.58; 89.31); 7.41-(127.01; 126.02; 89.31); 7.35-(145.53; 128.63); 7.31-(124.66; 128.5); 7.24-125.58; 7.20-126.02; (4.17; 3.76)-(89.31; 41.04; 28.52); 3.58-(64.59; 57.95); (2.24; 1.92)-(89.31; 64.59; 41.04).

Elemental analysis C₃₈H₄₈Cl₄N₂O₂Zn (M: 771.99) calculated: C, 59.12%; H, 6.27%; N, 3.63%; Cl, 18.37% measured: C, 58.73%; H, 6.40%; N, 3.57%; Cl, 18.82%

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Example 3

(R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1)
co-crystal (2c)



(R)-(+)-blarcamesine base (1.50 g, 5.33 mmol) was dissolved in IPA (10 mL), then warmed to 80° C. A pre-prepared 80° C. IPA solution of ZnCl₂ (0.40 g, 2.93 mmol, dissolved in 5 mL solvent) then a 6.25 M IPA/HCl solution (0.94 mL, 5.86 mmol, 1.1 mol eq.) was added. The mixture was stirred further, and the precipitated solid was filtered off still in warm. The solid crystalline material was dried in vacuum until constant weight, yielding 1.94 g (94.3%) white (R)-(+)-blarcamesine hydrochloride, ZnCl₂ (2:1) co-crystal.

HPLC purity: 99.73%

Mp.: 176-178° C.

Zn content (complexometric titration): 8.79% (calculated: 8.47%)

¹H-NMR (DMSO, 600 MHz): 9.93 (b, 1H, 7.62 (m, 2H); 7.42 (m, 2H); 7.35 (m, 2H); 7.31 (m, 2H); 7.24 (m, 1H); 7.20 (m, 1H); 4.18 (m, 1H); 3.77 (m, 1H); 3.59 (m, 1H); 2.83 (b, 3H); 2.74 (b, 3H); 2.72 (b, 2H); 2.26 (m, 1H); 1.93 (m, 1H).

¹³C-NMR (DMSO, 150 MHz): 145.50; 142.64; 128.63; 128.50; 127.25; 127.02; 126.01; 125.57; 89.33; 64.59; 57.90; 44.87.

COSY: 7.62-7.35-7.24; 7.42-7.31-7.20; (4.18; 3.77)-(2.26; 1.92)-3.59-2.72.

ROESY: 4.18-2.26; 3.59-(7.62; 1.92).

HSQC (140 Hz): 7.62-125.57; 7.42-126.01; 7.35-128.63; 7.31-128.50; 7.24-127.25; 7.20-127.02; 4.18-64.59; 3.77-64.59; 3.59-41.00; 2.83-41.34; 2.74-44.87; 2.72-57.90; 2.26-28.53; 1.93-28.53.

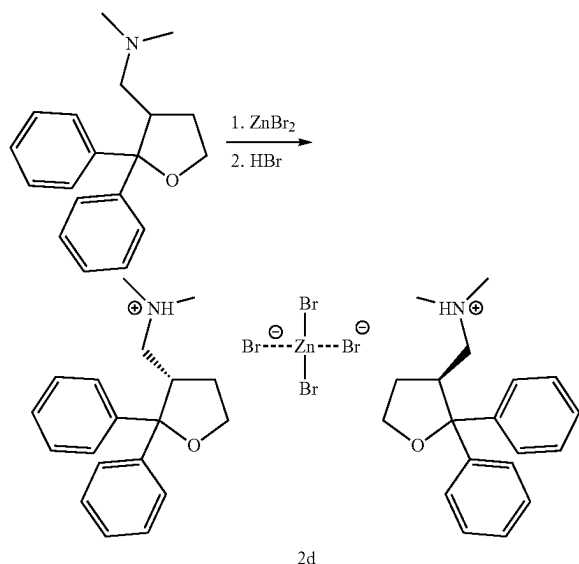
HMBC (8 Hz, 140 Hz): 7.62-(127.25; 125.57; 89.33); 7.42-(127.02; 126.01; 89.33); 7.35-(145.50; 128.63); 7.31-(142.64; 128.50); 7.24-125.57; 7.20-126.01; (4.18; 3.77)-(89.33; 41.00; 28.53); 3.59-(64.59; 57.90); 2.72-(41.00; 28.53); (2.26; 1.93)-(89.33; 64.59; 57.90; 41.00).

Elemental analysis C₃₈H₄₈Cl₄N₂O₂Zn (M: 771.99) calculated: C, 59.12%; H, 6.27%; N, 3.63%; Cl, 18.37% measured: C, 58.96%; H, 6.40%; N, 3.61%; Cl, 18.39%

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Example 4

(S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal (2d)



Using the method described in examples 2 and 3, 0.53 g of (S)-(-)-blarcamesine hydrobromide, (R)-(+)-blarcamesine hydrobromide, ZnBr_2 (1:1:1) co-crystal was prepared. Mp.: 181-183.5° C.

$^1\text{H-NMR}$ (DMSO, 600 MHz): 7.59 (m, 2H); 7.36 (m, 2H); 7.34 (m, 2H); 7.28 (m, 2H); 7.23 (m, 1H); 7.17 (m, 1H); 4.11 (m, 1H); 3.72 (m, 1H); 3.38 (b, 1H); 2.46 (b, 6H); 2.30 (b, 2H); 2.09 (m, 1H); 1.86 (m, 1H).

$^{13}\text{C-NMR}$ (DMSO, 150 MHz): 146.06; 143.19; 128.54; 128.22; 127.10; 126.75; 126.13; 125.75; 89.16; 64.68; 59.20; 44.69; 42.32; 28.59.

COSY: 7.59-7.34-7.23; 7.36-7.28-7.17; (4.11; 3.72)-(2.09; 1.86)-3.38.

HSQC (140 Hz): 7.59-125.75; 7.36-126.13; 7.34-128.54; 7.28-128.22; 7.23-127.10; 7.17-126.75; 4.11-64.68; 7.32-64.68; 3.38-42.32; 2.46-44.69; 2.30-59.20; 2.09-28.59; 1.86-28.59.

HMBC (8 Hz, 140 Hz): 7.59-(127.10; 125.75; 89.16); 7.36-(126.75; 126.13; 89.16); 7.34-(146.06; 128.54); 7.28-(143.19; 128.22); 7.23-125.75; 7.17-126.13; (4.11; 3.72)-(89.16; 42.32; 28.59); (2.09; 1.86)-(89.16; 64.65; 59.20; 42.32).

Elemental analysis $\text{C}_{38}\text{H}_{48}\text{Br}_4\text{N}_2\text{O}_2\text{Zn}$ (M: 949.81) calculated: C, 48.05%; H, 5.09%; N, 2.95%; Br, 33.65% measured: C, 45.52%; H, 4.71%; N, 2.79%; Br, 32.52%

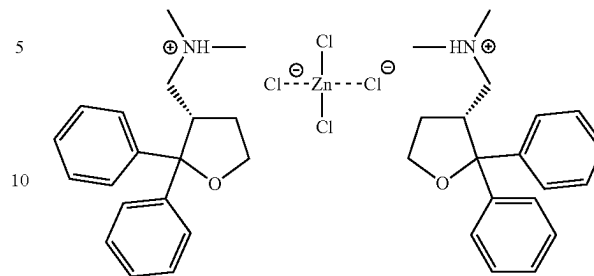
The invention claimed is:

1. A method for treating a neurodegenerative or neurodevelopmental disease, comprising administering to a subject in need thereof a therapeutically effective amount of a pharmaceutical dosage form comprising a therapeutically effective amount of a co-crystal of formula 2a, 2b or 2c and at least one therapeutically acceptable excipient,

wherein the co-crystal of formula 2a is a racemic co-crystal of formula 2a consisting of one molecule of (R) and one molecule of (S) configuration of blarcamesine hydrochloride and one molecule of ZnCl_2 per unit cell

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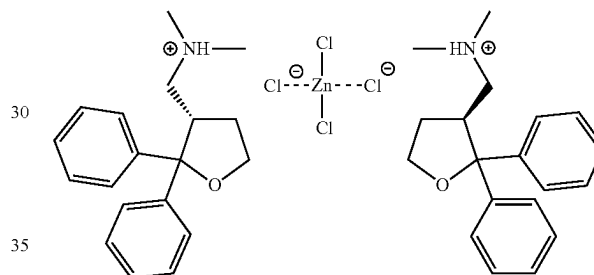
2a



characterized by an XRPD pattern having peaks at 7.93; 9.66; 13.73; 15.77; 16.73; and 17.82° $2\theta \pm 0.2^\circ$ 2θ ;

wherein the co-crystal of formula 2b is a co-crystal of formula 2b consisting of two molecules of (S) configuration of blarcamesine hydrochloride and one molecule of ZnCl_2 per unit cell

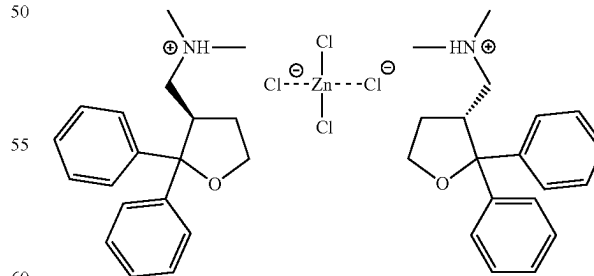
2b



characterized by an XRPD pattern having peaks at 12.15; 12.85; 13.04; 13.24; 16.36; and 19.67° $2\theta \pm 0.2^\circ$ 2θ ;

wherein the co-crystal of formula 2c is a co-crystal of formula 2c consisting of two molecules of (R) configuration of blarcamesine hydrochloride and one molecule of ZnCl_2 per unit cell

2c

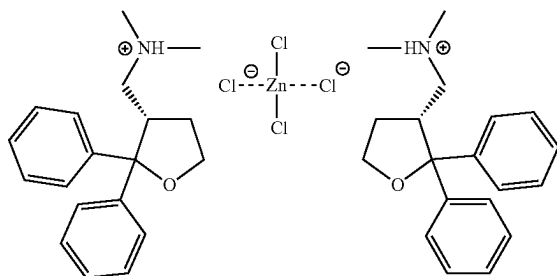


characterized by an XRPD pattern having peaks at 12.15; 12.85; 13.04; 13.24; 16.36; and 19.67° $2\theta \pm 0.2^\circ$ 2θ ; and

wherein the neurodegenerative or neurodevelopmental disease to be treated is Alzheimer's disease, dementia due to Parkinson's disease, or Rett syndrome.

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2. The method according to claim 1 wherein the co-crystal is the racemic co-crystal of formula 2a consisting of one molecule of (R) and one molecule of (S) configuration of blarcamesine hydrochloride and one molecule of ZnCl_2 per unit cell

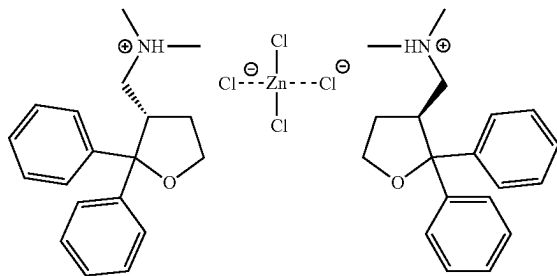


characterized by an XRPD pattern having peaks at 7.93; 9.66; 13.73; 15.77; 16.73; and 17.82° $2\theta \pm 0.2^\circ$ 2θ .

3. The method according to claim 2 wherein the co-crystal is characterized by the following data:

the N (+)-Cl (-) distance is 0.31-0.33 nm.

4. The method according to claim 1 wherein the co-crystal is the co-crystal of formula 2b consisting of two molecules of (S) configuration of blarcamesine hydrochloride and one molecule of ZnCl_2 per unit cell



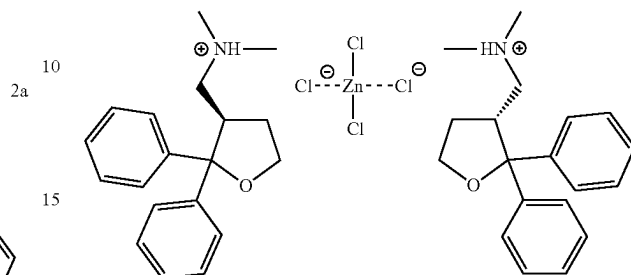
characterized by an XRPD pattern having peaks at 12.15; 12.85; 13.04; 13.24; 16.36; and 19.67° $2\theta \pm 0.2^\circ$ 2θ .

5. The method according to claim 4 wherein the co-crystal is characterized by the following data:

the N (+)-Cl (-) distance is 0.31-0.33 nm.

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6. The method according to claim 1 wherein the co-crystal is the co-crystal of formula 2c consisting of two molecules of (R) configuration of blarcamesine hydrochloride and one molecule of ZnCl_2 per unit cell



characterized by an XRPD pattern having peaks at 12.15; 12.85; 13.04; 13.24; 16.36; and 19.67° $2\theta \pm 0.2^\circ$ 2θ .

7. The method according to claim 6 wherein the co-crystal is characterized by the following data:

the N (+)-Cl (-) distance is 0.31-0.33 nm.

8. The method according to claim 1, wherein the neurodegenerative or neurodevelopmental disease to be treated is Alzheimer's disease.

9. The method according to claim 1, wherein the neurodegenerative or neurodevelopmental disease to be treated is dementia due to Parkinson's disease.

10. The method according to claim 1, wherein the neurodegenerative or neurodevelopmental disease to be treated is Rett syndrome.

11. The method according to claim 2, wherein the neurodegenerative or neurodevelopmental disease to be treated is Alzheimer's disease.

12. The method according to claim 2, wherein the neurodegenerative or neurodevelopmental disease to be treated is dementia due to Parkinson's disease.

13. The method according to claim 2, wherein the neurodegenerative or neurodevelopmental disease to be treated is Rett syndrome.

14. The method according to claim 4, wherein the neurodegenerative or neurodevelopmental disease to be treated is Alzheimer's disease.

15. The method according to claim 4, wherein the neurodegenerative or neurodevelopmental disease to be treated is dementia due to Parkinson's disease.

16. The method according to claim 4, wherein the neurodegenerative or neurodevelopmental disease to be treated is Rett syndrome.

17. The method according to claim 6, wherein the neurodegenerative or neurodevelopmental disease to be treated is Alzheimer's disease.

18. The method according to claim 6, wherein the neurodegenerative or neurodevelopmental disease to be treated is dementia due to Parkinson's disease.

19. The method according to claim 6, wherein the neurodegenerative or neurodevelopmental disease to be treated is Rett syndrome.

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